A Source-Explicit Conditional Common-Weight All-m Type-II Estimate and the Complementary Signed High-Order Frontier

Late 27 August V14: exact Vaughan affine-prime source and prime-modulus Möbius two-outer frontier

Research draft

August 27, 2026

Controlling V14 source update: exact Vaughan affine-prime source

Publication/status firewall. This update supersedes the older controlling-frontier labels in the underlying Draft 13 wherever they conflict. It does not alter the finite Lean theorem and it does not claim Erdős Problem #287 solved. The purpose is to replace the previously over-broad determinant-one source model by the literal source obtained from $\Lambda(2mn \pm 1)$ itself.

The controlling research ledger is now

`FORD-GENERATED-DEPTH-NO-287`	PASS, $N_0 = 112$,
fixed-g_* depth	$N_0 = 76$,
$k = 0$ smooth depth	40,
`AFFINE287-DET1-HYBRID-MQ45`	SUPERSEDED AS CONTROLLING SOURCE,
`AFFINE287-VAUGHAN-PRIME-SOURCE45`	PASS,
`AFFINE287-VAUGHAN-PRIMEPOWER45`	PASS,
`VAUGHAN-COFACTOR-WELLFACTORABLE45`	FALSE (literal no–go),
`SINGLETON-TO-GATE1A-DIRECT45`	FAIL,
`SINGLETON-TO-NATIVE-QK56-DIRECT45`	NO LITERAL DICTIONARY,
`AFFINE287-PRIME-MODULUS-MU-TWOOUTER45`	**FIRST TRUE ANALYTIC OPEN**,
`FCL`	OPEN,
Erdős #287	OPEN.

(V14.1)

1. Fixed-depth Ford source

Set

$$\nu_0 = \frac{16623}{100000}, \qquad 0 < \varepsilon_* < \frac{\nu_0}{100}, \qquad \sigma = \nu_0 - 2\varepsilon_*.$$

Then $\sigma > 0.98\nu_0 = 0.1629054$. The Ford–Maynard fragmentation bound gives at most $\lfloor 3/\sigma \rfloor + 2 \le 20$ pieces on each smooth side. For the general source, $k \le \lfloor 1/\sigma \rfloor \le 6$, while with $\gamma = 1/2 - \varepsilon_*$ one has

$$\ell = 6 \left\lceil \frac{1}{1-\gamma} \right\rceil = 12.$$

Hence a safe fixed depth is

$$\boxed{N = k\ell + r + s \le 6 \cdot 12 + 20 + 20 = 112.} \tag{V14.2}$$

For the pinned fixed certificate g_* the high-prime dimension is at most three, giving $N \le 76$; for the present $k = 0$ smooth packet, $N \le 40$. Thus no unbounded-convolution-depth premise is hidden in the source decomposition.

2. Exact affine Vaughan decomposition

For arithmetic functions define

$$\mu_{\le U}(d) = \mu(d)\mathbf{1}_{d \le U}, \qquad \mu_{>U} = \mu - \mu_{\le U},$$

$$\Lambda_{\le V}(e) = \Lambda(e)\mathbf{1}_{e \le V}, \qquad \Lambda_{>V} = \Lambda - \Lambda_{\le V}.$$

Using $\log = \Lambda * 1$ and $\mu * 1 = \varepsilon$, the exact identity is

$$\boxed{\Lambda = \Lambda_{\le V} + \mu_{\le U} * \log - \mu_{\le U} * \Lambda_{\le V} * 1 + \mu_{>U} * \Lambda_{>V} * 1.} \tag{V14.3}$$

For

$$L_s = 2mn + s, \qquad s \in \{-1, +1\},$$

and $U = V = X^{1/3}$, the large-X affine prime source therefore splits as

$$\Lambda(L_s) = \mathcal{I}_1(L_s) + \mathcal{I}_2(L_s) + \mathcal{II}(L_s), \tag{V14.4}$$

where

$$\mathcal{I}_1(L_s) = \sum_{\substack{d|L_s\\ d\le U}} \mu(d) \log \frac{L_s}{d}, \tag{V14.5}$$

$$\mathcal{I}_2(L_s) = -\sum_{\substack{der=L_s\\ d\le U,\ e\le V}} \mu(d)\Lambda(e), \tag{V14.6}$$

and

$$\boxed{\mathcal{II}(L_s) = \sum_{\substack{der=L_s\\ d>U,\ e>V}} \mu(d)\Lambda(e).} \tag{V14.7}$$

This is the literal affine-prime source. The historical shifted-Möbius/F3 object $\sum \alpha_m\beta_n\mu(mn+2)$ is not used as a surrogate for $\Lambda(2mn \pm 1)$.

3. Proper prime powers are negligible

In (V14.7), separate $e = p^j$, $j \ge 2$. Fixed source depth and divisor multiplicity cost only $X^{o(1)}$, so

$$\mathcal{E}_{\rm pp} \ll X^{o(1)} \sum_{\substack{p^j>V\\ j\ge 2}} \left(\frac{X}{p^j} + 1\right) \ll X^{5/6+o(1)}, \tag{V14.8}$$

for $V = X^{1/3}$, with higher powers smaller. Thus the proper-prime-power child has relative margin $X^{-1/6+o(1)}$ and is negligible on the $X/\log X$ scale.

4. The source-exact prime-outer packet

The hard part is $e = p$ prime. On dyadic cells

$$d \asymp D, \quad p \asymp P, \quad r \asymp R, \quad D, P > X^{1/3-o(1)}, \quad DPR \asymp X, \quad R < X^{1/3+o(1)},$$

with

$$MN \asymp X, \qquad X^{\sigma/3} < M \le X^{\sigma} < X^{1/6},$$

the literal source is

$$\boxed{\begin{aligned}\mathcal{V}_{\pi,s}(D,P,R) = &\sum_{\substack{m\sim M,\ n\sim N\\ d\sim D,\ p\sim P,\ r\sim R\\ dpr=2mn+s}} \xi_\pi(m)\kappa_\pi(n)\mu(d)(\log p)W_{\pi,s}(m,n,d,p,r)\\ &- \mathfrak{M}_{\pi,s}(D,P,R).\end{aligned}} \tag{V14.9}$$

Here $\mathfrak{M}_{\pi,s}$ is the comparison-sequence local-density packet derived from the same source partition; it is not an independently chosen normalizing constant.

Folding $q = dr$, define

$$\lambda_U(q) = \sum_{\substack{dr=q\\ d>U}} \mu(d). \tag{V14.10}$$

Then

$$\boxed{\mathcal{V}_{\pi,s}(Q,P) = \sum_{\substack{m\sim M,\ n\sim N\\ q\sim Q,\ p\sim P\\ qp=2mn+s}} \xi_\pi(m)\kappa_\pi(n)\lambda_U(q)(\log p)W_{\pi,s} - \mathfrak{M}_{\pi,s}(Q,P),} \tag{V14.11}$$

where $QP \asymp X$ and $X^{1/3} \ll Q, P \ll X^{2/3}$. Since $\sum_{d|q}\mu(d) = 0$ for $q > 1$,

$$\boxed{\lambda_U(q) = -\sum_{\substack{d|q\\ d\le U}} \mu(d) \qquad (q > U).} \tag{V14.12}$$

Thus the modulus/cofactor weight is a coherent truncated Möbius divisor sum, not an arbitrary divisor-bounded weight.

5. Determinant-one prime parametrisation

The equation

$$qp - 2mn = s \tag{V14.13}$$

implies

$$(q, 2m) = (p, n) = (q, n) = (p, m) = 1. \tag{V14.14}$$

For fixed q, m, choose one solution $qp_0 - 2mn_0 = s$. Then every integer solution is

$$\boxed{p = p_0 + 2mt, \qquad n = n_0 + qt.} \tag{V14.15}$$

Hence the genuine analytic content is a source-matched version of

$$\boxed{\sum_{q,m} \lambda_U(q)\xi_\pi(m) \sum_{t\in I_{q,m}} \log(p_0 + 2mt)\, \mathbf{1}_{p_0+2mt \text{ prime}}\, \kappa_\pi(n_0 + qt),} \tag{V14.16}$$

minus its literal comparison main term.

6. Prime-modulus Möbius two-outer orientation

Keeping p as modulus gives the source-exact two-outer packet

$$\boxed{\mathcal{R}^{(2)}_{\pi,s}(P; D, R) = \sum_{p\sim P}(\log p) \sum_{\substack{m\sim M,\ n\sim N\\ 2mn\equiv -s \pmod p}} \xi_\pi(m)\kappa_\pi(n)\, \Delta^{\mu,1}_{D,R}\left(\frac{2mn+s}{p}\right) - \mathfrak{M}_{\pi,s},} \tag{V14.17}$$

where

$$\boxed{\Delta^{\mu,1}_{D,R}(u) = \sum_{\substack{dr=u\\ d\sim D,\ r\sim R}} \mu(d)W_D(d)W_R(r).} \tag{V14.18}$$

The modulus p is prime, the first quotient outer is a Möbius block, the second is a model block, and the affine residue is the fixed unit $-s\overline{2} \pmod p$.

This establishes a structural intersection with the Gate-1B two-outer/F3 frontier, but not a pre-existing closed provider theorem. The current analytic target is therefore

$$\boxed{\texttt{AFFINE287-PRIME-MODULUS-MU-TWOOUTER45.}} \tag{V14.19}$$

A sufficient estimate is

$$\boxed{|\mathcal{R}^{(2)}_{\pi,s}| = o(X/\log X),} \tag{V14.20}$$

or a sufficiently small fixed constant multiple compatible with the fixed-certificate positivity margin. The presently available general estimate is only $X(\log X)^{O(1)}$.

7. Two exact provider no-go statements

First, λ_U is not a well-factorable modulus weight in the required literal sense. For a prime $\ell > U$, $\lambda_U(\ell) = \mu(\ell) = -1$. A dyadic well-factorable decomposition at a balanced level would force the convolution to vanish at primes in the top dyadic interval because neither short factor can support ℓ. Thus

$$\boxed{\texttt{VAUGHAN-COFACTOR-WELLFACTORABLE45:}\quad \texttt{FALSE.}} \tag{V14.21}$$

Consequently well-factorable AP theorems do not directly fill the actual λ_U socket.

Second, the canonical Gate-1A row has determinant $2k$ with $k \in \mathbb{Z}$, whereas the affine Vaughan packet has determinant $s = \pm 1$. A direct identification would require $2k = s$, hence $k = \pm 1/2$. Therefore

$$\boxed{\texttt{SINGLETON-TO-GATE1A-COMMONWEIGHT: DIRECT LITERAL ADAPTER FAILS.}} \tag{V14.22}$$

This excludes only the existing canonical Gate-1A dictionary, not a future odd-determinant variant.

8. Remaining sibling/source obligations

The Vaughan "Type-I" siblings (V14.5)–(V14.6) still carry the generated quotient coefficient $\kappa_\pi(n)$, so their name does not make them automatic consequences of the earlier Gate-0 Type-I theorem. The first source/standard adapter pin remains

$$\boxed{\texttt{VAUGHAN-TYPEI-GENERATED-KAPPA45.}} \tag{V14.23}$$

Comparison-side Vaughan matching is likewise a source pin. Direct native QK56 identification has not been established.

9. Current closure graph

$$\boxed{\begin{array}{c}
\text{fixed } g_* + k = 0 \text{ leakage} + \text{singleton fragmentation} \quad \textbf{PASS} \\
\Downarrow \\
\text{exact affine Vaughan decomposition} \quad \textbf{PASS} \\
\Downarrow \\
\left\{\begin{array}{ll}
\text{proper prime-power outer} & \textbf{NEGLIGIBLE}, \\
\text{Vaughan Type-I generated-}\kappa & \textbf{SOURCE/STANDARD OPEN}, \\
\text{prime outer } p + \mu\text{-quotient} & \textbf{FRONTIER},
\end{array}\right. \\
\Downarrow \\
\boxed{\texttt{AFFINE287-PRIME-MODULUS-MU-TWOOUTER45}} \\
\Downarrow \\
k = 0 \text{ smooth-parity packet closed} \\
\Downarrow \\
\text{remaining actual fixed-}g_* \text{ packets} \\
\Downarrow \\
\texttt{FCL-CLOSED} + \text{comparison}/N_2/\varepsilon_* \\
\Downarrow \\
\text{positive affine prime mass} \\
\Downarrow \\
\texttt{WindowPairSupply}/\text{finite Lean compiler} \\
\Downarrow \\
\text{Erdős Problem \#287.}
\end{array}} \tag{V14.24}$$

Controlling verdict. The first source-exact analytic obstruction is no longer a vague generated Type-II theorem and no longer the over-broad generic determinant hybrid. It is the prime-modulus, one-Möbius-outer, one-model-outer affine packet `AFFINE287-PRIME-MODULUS-MU-TWOOUTER45`. The $k = 0$ packet, FCL, the large-M bridge and Erdős #287 remain open.

Abstract

This note records an audited public-facing state of a parity-sensitive Type-II programme. Part I studies a canonical common-weight direct operator. A row is indexed by a prime $r \asymp R$, a shift k, and $m \asymp M$, with $m' = m + kr \asymp M$. Under five explicit source/analytic pins—one-sided Poisson–Bruhat transport and an absolute source ceiling, fixed-state exclusion, the outer-root energy pin, projective source routing, and the literal recombination estimate—band-limited prime participation yields the required $R^{-1/4+o(1)}$ contraction. The finite participation inequality, projective ratio-class algebra, exponent ledger, and normalization compilers are separately machine-checked; the displayed source/analytic pins are not silently promoted into Lean theorems.

Part II records the complementary Gate-1B frontier. Returning to the native order-five shell and opening the first TT^* covariance gives an exact sector law: the same-modulus branch is a character Gram, the coprime cross-modulus branch is a CRT Kloosterman family, and the shared-divisor branch belongs to the existing GCD router. The proper smooth same-q cross-character child and the coprime cross-q no-wrap operator remain useful positive analytic children. The later high-conductor audit first recovered a signed primitive-conductor parent by writing $q = cr$: the induced Gauss square preserves one complement factor $\mu(r)$, forcing any sign-based high-conductor repair to recombine same-q with common-conductor cross-q covariance. A subsequent hostile audit then shows that this common-conductor square norm is not the full parity-breaking parent. It squares away the conductor sign $\mu(c)$; in particular the near-primitive slice $r = 1$ retains no complement sign at all. The same audit proves an exact primitive-Kloosterman inversion and shows that the naive square-complement Pascadi dictionary is false: the physical inverse variable is r^2B, not r^2, the two genuine smooth slots are already occupied by the dual frequencies h, k, and the two legal Proposition 3.8 dictionaries are fixed-power nonclosing. The common-conductor Pascadi closure route is therefore retired. The present first analytic open is the near-primitive full-conductor five-defect signed packet isolated in Late update V. The actual switched c_9 dictionary and the literal shifted pre-enlargement TT^* source also remain source open. The canonical Gate-1B additive zero mode remains an exact finite/algebraic pass in the V10 Lean bank.

Part III records a refined Ford–Maynard landing strip. Ford–Maynard Theorem 8.3 gives a published five-collar description of the exceptional H_2 geometry, and Definition 5.1 identifies the coordinates of $\mathbf{v}(n)$ with logarithms of the individual prime factors. This permits a more source-specific treatment of the rough N_2 sector: squareful terms are negligible, while the squarefree shifted-prime lane reduces, at the research level, to a standard uniform upper sieve for two linear forms plus the published rough-number geometry. A proof audit of Proposition 7.22 shows that the published universal Type-II hypothesis is invoked only after the proof has generated the grouped equation (7.23), so controlling every *actual generated* instance of that structured sum is a proof-specific sufficient landing point. Internal Ford-side archaeology further indicates finite-depth generated one-variable factors and finite nuclear/weight-dependence compilers; however, the literal source-exact weighted high-complexity packet dictionary from Gate outputs to the Ford-generated factors remains missing. We call this source interface `SOURCE-ADAPTER45`. These are programme reductions, not weaker theorems published by Ford–Maynard.

No claim of Gate-1B closure, Full Type-II reassembly, a Hardy–Littlewood prime-pair asymptotic, or twin-prime infinitude is made.

Contents

1 Introduction

A recurring obstruction in parity-breaking sieve arguments is that strong fixed-slice estimates need not survive coherent averaging over the true source family. The correct object is therefore the source-restricted family operator, not a collection of unrelated local norms.

This note isolates a canonical direct Type-II operator and derives a conditional all-m mean-square bound under explicit source and analytic pins. Four structural ingredients drive the proof:

(i) exact source and determinant-2 row geometry;

(ii) source-preserving Fourier/Poisson transport to a smooth moving-r family;

(iii) band-limited prime participation, converting smoothness in r into a moving-family contraction;

(iv) an exact projective crossed-convolution identity for the zero-curvature branch.

The downstream motivation is the Ford–Maynard prime-producing sieve framework, where Type I and Type II information for an actual non-negative sequence is used simultaneously [FM24]. Their work also shows that a substantial Type II range is genuinely necessary for non-trivial lower bounds. We do not perform the full downstream reassembly here.

1.1 External analytic input

The participation step needs a prime number theorem in intervals of length $R^{3/4}$. Huxley's classical short-interval prime number theorem gives, uniformly for $x \asymp R$,

$$\pi(x+H) - \pi(x) = (1+o(1))\frac{H}{\log x} \qquad (H \geq x^{7/12+\varepsilon}),$$

for every fixed $\varepsilon > 0$ [Hux72]. Since $3/4 > 7/12$, the plateaux used below are longer than the range required by Huxley's theorem, so no modern strengthening is needed. We also use the classical Bernstein inequality

$$\|P'\|_\infty \leq N\|P\|_\infty$$

for trigonometric polynomials of degree at most N [Zyg02].

1.2 Scope

The theorem proved below concerns the canonical *common-weight* direct source. In the authoritative source, the physical smoothing datum is one weight W_D; row dependence that appears later is derived deterministically from this common datum through Fourier, reciprocity and Poisson transforms. An abstract interface allowing arbitrary independent edge-dependent weights is strictly more general and belongs to an upstream source-adaptation problem. Likewise, a global packet census and Full Type II reassembly are not part of the theorem proved here. This distinction is also machine-checked in the formal development: the Gate-1A direct closure certificate contains no global packet-dictionary or edge-dependent-D2 field.

2 Parameters and notation

Let $X \to \infty$. Set

$$M = X^{1/3}, \qquad R = X^a, \qquad L = X^b,$$

and

$$H = X^{a+2b-2/3}, \qquad K = \frac{M}{R}, \qquad U = \frac{L}{H}, \qquad D = \frac{L^2}{H}.$$

We repeatedly use

$$RK = M, \qquad DH = L^2, \qquad M^2H = RL^2. \tag{1}$$

The three controlling vertices are

$$V_1 = \left(\frac{5}{18}, \frac{1}{3}\right), \qquad V_2 = \left(\frac{5}{18}, \frac{25}{72}\right), \qquad V_3 = \left(\frac{7}{24}, \frac{1}{3}\right).$$

For a prime ℓ, define

$$\rho_\ell(n) := \mathbf{1}_{\ell | n} - \frac{1}{\ell}. \tag{2}$$

Let b_p, d_q be 1-bounded coefficient sequences on primes $p, q \asymp L$. All harmless divisor and logarithmic losses are absorbed in $X^{o(1)}$.

We use $e(z) = e^{2\pi i z}$ and $e_c(z) = e(z/c)$. For a Schwartz function f we use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e(-x\xi)\, dx.$$

Every inverse symbol in a modular phase is understood only on the explicitly routed unit/full-conductor sector.

3 The canonical physical source

3.1 Rows and affine forms

A canonical row is $e = (r, k, m)$ with

$$r \asymp R \text{ prime}, \qquad 1 \le |k| \le K, \qquad m \asymp M, \qquad m' := m + kr \asymp M,$$

together with the generic unit/full-conductor conditions needed by the reciprocal phases. Let $\mathcal{E}^\sharp$ be this row family.

Choose w_0 modulo r such that

$$mw_0 \equiv -2 \pmod{r}.$$

Since $m' = m + kr \equiv m \pmod{r}$, the same choice satisfies $m'w_0 \equiv -2 \pmod{r}$. Hence

$$\alpha := \frac{mw_0 + 2}{r}, \qquad \beta := \frac{m'w_0 + 2}{r}$$

are integers. Define

$$A_e(t) := \alpha + mt, \qquad B_e(t) := \beta + m't.$$

Lemma 3.1 (Determinant-2 row identity)**.** *For every canonical row and every integer t,*

$$m'A_e(t) - mB_e(t) = 2k.$$

Proof. The t-terms cancel and

$$m'\alpha - m\beta = \frac{m'(mw_0 + 2) - m(m'w_0 + 2)}{r} = \frac{2(m' - m)}{r} = 2k.$$

$\square$

3.2 Common physical smoothing

Fix $W \in C_c^\infty(\mathbb{R})$ and write $W_D(t) = W(t/D)$. The key point is that this weight is common to all rows. Define

$$P_e(t) = \sum_{p \asymp L} b_p \rho_p(A_e(t)), \qquad Q_e(t) = \sum_{q \asymp L} d_q \rho_q(B_e(t)),$$

and

$$C_e(b, d) := \sum_{t \in \mathbb{Z}} W_D(t) P_e(t) Q_e(t). \tag{3}$$

Remark 3.2. An arbitrary family $W_{D,e}$ is not part of the canonical theorem. Such weights arise only in a more general source-adaptation interface.

3.3 Source provenance and derived row dependence

The common-weight assertion is stronger than a notational convention. After Fourier transformation the source coefficient factors as

$$\omega_{m,p,q}(h) = \underbrace{\frac{H}{pq} \mathrm{e}\left(-\frac{h\alpha}{pqm}\right)}_{\text{derived row-dependent scalar}} \underbrace{\widehat{W}_D\left(\frac{h}{pq}\right)}_{\text{one common physical weight}}, \tag{4}$$

where the argument $h/(pq)$ has no row index. Thus the later dependence on m, r, p, q is *derived dependence*; it is not freedom to choose an independent physical weight on each row. This distinction is load-bearing: an arbitrary edge-dependent family would be a strictly stronger source class and is deliberately excluded from the canonical direct theorem.

4 Fourier normalization and determinant shell

The normalized pre-square source is

$$\widetilde{C}_{r,k,m} = \sum_{p \asymp L} b_p \sum_{\substack{q \asymp L \\ q \neq p}} d_q \sum_h \omega_{m,p,q}(h) \mathrm{e}_{mq}\Big(2hk\overline{p(m+kr)}\Big), \tag{5}$$

where

$$\omega_{m,p,q}(h) = \frac{H}{pq} \widehat{W}_D\left(\frac{h}{pq}\right) \mathrm{e}\left(-\frac{h\alpha}{pqm}\right). \tag{6}$$

The source norms satisfy

$$\sum_h |\omega_{m,p,q}(h)|^2 \ll H, \qquad \sum_h |\omega_{m,p,q}(h)| \ll H.$$

Writing $m' = m + rk$, additive reciprocity separates the phase:

$$\mathrm{e}_{mq}\big(2hk\overline{pm'}\big) = \mathrm{e}_{pr}\big(-2h\overline{qm}\big) \, \mathrm{e}\left(\frac{2h}{prqm}\right) \mathrm{e}_q\big(-2h\overline{prm'}\big). \tag{7}$$

Thus the exact H-scale smooth profile is

$$\beta_{m,r;p,q}(h/H) = \frac{H}{pq} \widehat{W}_D(h/pq) \mathrm{e}\left(-\frac{h\alpha}{pqm} + \frac{2h}{prqm}\right).$$

The h-Poisson/Gauss completion gives the determinant shell

$$ms - prn = 2. \tag{8}$$

Choose $\kappa_p(m) \in [0, pr)$ with $m\kappa_p(m) \equiv 2 \pmod{pr}$ and set

$$n_0(m,p,r) = \frac{m\kappa_p(m) - 2}{pr}.$$

Then every solution is parameterized by

$$s_j = \kappa_p(m) + jpr, \qquad n_j = n_0 + jm.$$

The quotient amplitude is

$$\mathcal{W}_{m,r;p,q}(j) := \widehat{\beta}_{m,r;p,q}\left(\frac{H(n_0 + jm)}{qm}\right), \tag{9}$$

and, for $|j| \ll U = L/H$,

$$\sum_{|j| \ll U} |\mathcal{W}_{m,r;p,q}(j)|^2 \ll UX^{o(1)}. \tag{10}$$

5 Source-preserving Poisson–Bruhat transport

The following source-specific statement is isolated because it is the exact bridge between the authoritative physical source and the finite BPP compiler. The scale identity in Lemma 5.3 is elementary; the uniform smooth envelope, the negligible-state disposal, and the numerical source ceiling are source/analytic inputs.

For a transported state family put

$$a_{r,\xi} := |c_r(\xi)|, \qquad \mathcal{T}_{\mathrm{abs}} := \sum_{\substack{r \in [R,2R] \\ r \text{ prime}}} \left(\sum_\xi a_{r,\xi}\right)^2.$$

Hypothesis 5.1 (One-sided source transport and absolute ceiling)**.** *For every fixed $A_0 > 0$, the transported source splits into a retained finite state set $\mathfrak{X}_{\mathrm{ret}}$ and a discarded remainder with Gate-energy contribution $O_{A_0}(X^{-A_0})$ times the natural scale. Every retained state has scalar moving-row amplitude*

$$c_r(\xi) = B_\xi F_\xi(r/R), \tag{11}$$

where B_ξ is independent of r, F_ξ is supported in a fixed compact subinterval of $(1,2)$, and for every fixed J,

$$\sup_\xi \sup_t |\partial_t^J F_\xi(t)| \ll_J X^{o(1)}.$$

For every fixed truncation exponent A, the retained set is chosen so that $\|F_\xi\|_\infty \geq 10R^{-A}$; states failing this inequality are part of the discarded source remainder. The transport is one-sided, has $X^{o(1)}$ nuclear cost, and gives the explicit absolute source ceiling

$$\boxed{\mathcal{T}_{\mathrm{abs}} \ll M^2 L^4 X^{o(1)}.} \tag{12}$$

No two-sided Hilbert–Schmidt equivalence is assumed. Finally, the physical and normalized sources obey

$$C_e = H^{-1}\widetilde{C}_e + \mathcal{R}_e, \qquad \sum_{e\in\mathcal{E}^\sharp} |\mathcal{R}_e|^2 \ll_{A_0} X^{-A_0}\frac{ML^4}{H},$$

for every fixed $A_0 > 0$.

Status note 5.2 (Source status)**.** The finite Poisson–Bruhat algebra and the budget identity below are banked. The full assertion in Hypothesis 5.1, including the literal source equality, uniform retained-state truncation, the physical-normalized bridge and (12), is an explicit source/analytic pin. It is not inferred from the exponent arithmetic.

Lemma 5.3 (Poisson–Bruhat budget identity)**.** *The scale identities imply*

$$\boxed{\frac{MK(MH)^2}{RL^4} = 1.}$$

Proof. By (1), $MK = L^2/H$. Therefore

$$\frac{MK(MH)^2}{RL^4} = \frac{(L^2/H)M^2H^2}{RL^4} = \frac{M^2H}{RL^2} = 1.$$

□

Remark 5.4. The amplitude is smooth in r but is not constant. Thus a count of the prime divisors of a fixed determinant does not by itself give an R^{-1} weighted family saving.

6 Band-limited prime participation

From now on ξ ranges over the retained state set of Hypothesis 5.1. Put $a_{r,\xi} = 0$ when the prime row r is excluded for the state ξ, and define

$$M_\xi := |B_\xi| \sup_{1\leq t\leq 2} |F_\xi(t)|.$$

Fix a large A and put $N = R^{1/4}$. A smooth periodic extension of F_ξ and repeated integration by parts give, uniformly in retained ξ,

$$F_\xi(t) = P_\xi(t) + O_A(R^{-A}), \tag{13}$$

where P_ξ is a trigonometric polynomial of degree at most N.

Lemma 6.1 (Bernstein plateau)**.** *For every retained state ξ, there is an interval $I_\xi \subset [R, 2R]$ of length $\gg R^{3/4}$ on which $|c_r(\xi)| \gg M_\xi$.*

Proof. Let t_ξ maximize $|P_\xi|$. Bernstein's inequality [Zyg02] gives $\|P'_\xi\|_\infty \le N\|P_\xi\|_\infty$. Hence on a one-sided or two-sided interval of t-length $\gg N^{-1}$ contained in $[1, 2]$, P_ξ retains a fixed proportion of its maximum. The retained-state clause in Hypothesis 5.1 makes the error in (13) smaller than that maximum. Multiplying the interval length by R gives $R/N = R^{3/4}$. □

Hypothesis 6.2 (Fixed-state exclusion)**.** *For each retained state ξ, every prime row excluded from the generic non-projective branch divides one of $X^{o(1)}$ fixed nonzero integers $N_{\xi,j}$ of polynomial height, $|N_{\xi,j}| \le X^{O(1)}$.*

Lemma 6.3 (Admissible prime participation)**.** *Assume Hypothesis 6.2. For every retained state ξ, the plateau I_ξ contains $R^{3/4-o(1)}$ admissible prime rows. In particular, if*

$$A_\xi := \max_{\substack{r\sim R \\ r \text{ prime}}} a_{r,\xi}, \qquad N_\xi := \frac{\sum_{r\sim R,\ r \text{ prime}} a_{r,\xi}}{A_\xi},$$

then $A_\xi > 0$ and

$$A_\xi \asymp M_\xi, \qquad N_\xi \gg R^{3/4-o(1)}. \tag{14}$$

Proof. Huxley's short-interval prime number theorem applies to the interval from Lemma 6.1, since $3/4 > 7/12 + \varepsilon$ for a fixed small $\varepsilon > 0$ [Hux72]. Thus I_ξ contains $R^{3/4-o(1)}$ primes. A nonzero polynomial-height integer has $O(\log X)$ distinct prime divisors, and only $X^{o(1)}$ such integers occur, so the exclusion removes only $X^{o(1)}$ primes. On every remaining plateau prime, $a_{r,\xi} \gg M_\xi$, proving both claims. □

Let $\mathcal{R}$ be the primes in $[R, 2R]$, let $S = |\mathcal{R}| = R^{1+o(1)}$, and retain the notation $\mathcal{T}_{\text{abs}}$ from Section 5.

Lemma 6.4 (Participation-to-envelope). *If $N_\xi \ge P$ for every retained state, then*

$$\left(\sum_\xi A_\xi\right)^2 \le \frac{S}{P^2}\mathcal{T}_{\text{abs}}.$$

Proof. By Cauchy–Schwarz,

$$\left(\sum_\xi A_\xi N_\xi\right)^2 = \left(\sum_{r\in\mathcal{R}}\sum_\xi a_{r,\xi}\right)^2 \le S\mathcal{T}_{\text{abs}}.$$

Since $N_\xi \ge P$, the result follows. □

6.1 Projective versus non-projective Gram pairs

The transport attaches to every retained state a pair (Z_ξ, L_ξ) and the outer determinant

$$\Delta(\xi,\eta) := Z_\xi L_\eta - Z_\eta L_\xi. \tag{15}$$

We split the Gram pair set before applying BPP:

$$\Delta(\xi,\eta) \ne 0 \qquad \text{or} \qquad \Delta(\xi,\eta) = 0.$$

The second class is the projective branch in Section 8. Since $\Delta(\xi,\xi) = 0$, the entire state diagonal lies in that branch.

For $\Delta(\xi,\eta) \neq 0$, a row prime can couple the fixed pair only if it divides the fixed nonzero polynomial-height integer $\Delta(\xi,\eta)$. Hence

$$\mathcal{E}_{\mathrm{np}} \ll X^{o(1)} \left(\sum_{\xi} A_{\xi} \right)^2 . \tag{16}$$

Proposition 6.5 (BPP family-energy contraction)**.** *Under Hypotheses 5.1 and 6.2,*

$$\boxed{\mathcal{E}_{\mathrm{np}} \ll R^{-1/2+o(1)} \mathcal{T}_{\mathrm{abs}}.}$$

In particular, by (12),

$$\mathcal{E}_{\mathrm{np}} \ll R^{-1/2+o(1)} M^2 L^4.$$

Proof. Use Lemma 6.4 with $P = R^{3/4-o(1)}$ and $S = R^{1+o(1)}$ in (16). □

7 The unique outer square root

The finite outer four-cycle inequality gives a single square root, but its application to the canonical Gate energy requires an exact energy pin. We state that pin explicitly.

Hypothesis 7.1 (Outer-root energy pin)**.** *Let $\mathcal{G}_{\mathrm{np}}$ denote the normalized Gate contribution of the non-projective transported branch. The source identification and the finite outer-four-cycle compiler give*

$$\mathcal{G}_{\mathrm{np}} \ll M^2 L^4 X^{o(1)} \left(\frac{\mathcal{E}_{\mathrm{np}}}{\mathcal{T}_{\mathrm{abs}}} \right)^{1/2} . \tag{17}$$

The finite square-root algebra behind (17) is

$$\sum_r \|A_r\|_{\mathrm{op}} \leq |\mathcal{R}|^{1/2} \left(\sum_r \mathcal{T}_r \right)^{1/2} ,$$

and is machine-checked. The equality identifying its abstract energy with the canonical Gate energy is the source pin in Hypothesis 7.1. Combining Proposition 6.5 and Hypothesis 7.1 gives

$$\mathcal{G}_{\mathrm{np}} \ll R^{-1/4+o(1)} M^2 L^4.$$

The required comparison is $MR^{-1/4} \leq H$, equivalently

$$\frac{5}{4}a + 2b - 1 \geq 0.$$

At V_1, V_2, V_3 the margins are respectively

$$\frac{1}{72}, \qquad \frac{1}{24}, \qquad \frac{1}{32}.$$

8 Projective reassembly: ratio classes

The complementary branch is $\Delta(\xi,\eta) = 0$, including $\xi = \eta$. On $L \neq 0$ the correct class is the projective ratio $\kappa(Z,L) = Z/L$, not the product ZL. For a finite transported state support $\mathcal{S}$, put $v_{Z,L} = A_Z \otimes B_L$ and

$$C_{\xi} := \sum_{(Z,L)\in\mathcal{S}:\ \kappa(Z,L)=\xi} v_{Z,L}.$$

Then the exact class-pushforward identity is

$$\sum_{\xi} \|C_{\xi}\|^2 = \sum_{(Z_1,L_1),(Z_2,L_2)\in\mathcal{S}} \mathbf{1}_{Z_1 L_2 = Z_2 L_1} \operatorname{Re}\langle v_{Z_1,L_1}, v_{Z_2,L_2} \rangle. \tag{18}$$

Hypothesis 8.1 (Projective source routing)**.** *The transported $\Delta = 0$ source is routed into positive primitive ratio-class packets of class multiplicity $X^{o(1)}$. One-zero axes and non-primitive/rank-loss states are routed to explicit exceptional blocks. Moreover the diagonal/projective source mass satisfies*

$$\sum_{(Z,L)\in\mathcal{S}} \|A_Z\|^2\|B_L\|^2 \ll MHL^4X^{o(1)}, \tag{19}$$

and every exceptional block is bounded at the same target.

Under Hypothesis 8.1, Cauchy within each ratio class and (18) give

$$\mathcal{G}_{\text{proj}} \ll MHL^4X^{o(1)}.$$

The finite ratio-class identity, primitive rigidity and multiplicity lemma are machine-checked; Hypothesis 8.1 is the physical source-routing pin.

9 Recombination error

Hypothesis 9.1 (Literal source recombination estimate)**.** *The quotient-cell/source recombination error obeys*

$$\mathcal{E}_{\text{rec}} \ll U^{-2+o(1)}M^2L^4. \tag{20}$$

Since $U = L/H$ and $DH = L^2$,

$$\frac{U^{-2}M^2L^4}{MHL^4} = \frac{M}{D}.$$

The margins at V_1, V_2, V_3 are $1/18, 1/18, 1/24$.

10 Conditional main theorem

Theorem 10.1 (Canonical all-m common-weight estimate)**.** *Let the parameters lie in the polytope generated by V_1, V_2, V_3. Let $\mathcal{E}^\sharp$ be the generic unit/full-conductor all-m row family and let $C_e(b,d)$ be the covariance* (3)*. Assume Hypotheses 5.1, 6.2, 7.1, 8.1 and 9.1. Then*

$$\boxed{\sum_{e\in\mathcal{E}^\sharp} |C_e(b,d)|^2 \ll \frac{ML^4}{H}X^{o(1)}.}$$

Equivalently,

$$\boxed{\sum_{e\in\mathcal{E}^\sharp} |\widetilde{C}_e(b,d)|^2 \ll MHL^4X^{o(1)}.}$$

Proof. The non-projective branch is bounded by Proposition 6.5 and Hypothesis 7.1. The complete $\Delta = 0$ branch, including the state diagonal, is bounded by Hypothesis 8.1 and the exact ratio-class identity. The discarded source and the recombination error are controlled by Hypotheses 5.1 and 9.1. These branches are exhaustive and disjoint. The normalized estimate follows. The physical-normalized identity in Hypothesis 5.1, together with its square-summable remainder, gives the physical estimate. □

Corollary 10.2 (Clean-P_3 restriction)**.** *Every clean-P_3 subfamily of $\mathcal{E}^\sharp$ satisfies the same bound.*

Proof. The energy is a sum of non-negative row energies. □

11 Scope separation: Gate 1A versus source routing

The theorem above is a conditional analytic theorem for the canonical direct operator. It does not assert that every packet in a global sieve decomposition has already been identified with this operator.

Suppose an upstream source compiler gives an exact decomposition

$$\mathcal{S}_{\mathrm{actual}} = \sum_{\lambda \in \Lambda_{\mathrm{gen}}} a_\lambda D_\lambda U_\lambda \mathcal{S}^\sharp_\lambda + \mathcal{S}_{\mathrm{exc}},$$

where U_λ is unitary/permutational and

$$\sum_\lambda |a_\lambda| \|D_\lambda\|_{\mathrm{op}} \le X^{o(1)}.$$

Then the main theorem controls the generic packets by Minkowski, provided each packet is truly an instance of the canonical operator. Proving the source identity, packet multiplicity, and exceptional routing is a separate Gate-0 compiler problem.

In particular, a common physical weight routes directly into the theorem. A fixed-dimensional smooth family generated deterministically from the common source can be decomposed into $X^{o(1)}$ common templates. By contrast, an arbitrary independently varying edge weight is a genuinely stronger object. In the formal development this more general datum is retained as an upstream Gate-0 source-adapter obligation; it is not a premise of the canonical Gate-1A direct theorem.

This separation matters downstream. The Ford–Maynard framework applies to the actual non-negative sequence and its complete Type-I/Type-II information [FM24]; a convenient analytic model cannot replace an unverified source census.

12 Formal verification and provenance

The accompanying Lean bank deliberately separates finite mathematics, source provenance, and deep analytic input. This is useful here because a theorem compiler can be perfectly correct while its external analytic hypothesis is not itself formalized in Mathlib.

12.1 Machine-checked finite/source skeleton

The v9.8 formal bank verifies, without user axioms or admitted proofs:

- the canonical pre-square direct source as a literal source transcription, together with the fact that the physical transform $\widehat{W}_D$ is one common datum;
- the all-m generic row type, with no clean-P_3 field;
- the exact normalized and physical Gate-1A energies and their H^2 normalization bridge;
- the finite participation-to-family-energy inequality;
- the exponent conversion $R^{-1/2} \mapsto R^{-1/4}$ and the margins $1/72, 1/24, 1/32$;
- the projective/axis finite algebra and the U^{-2} exponent budget;
- the logical separation between the Gate-1A direct theorem and the upstream global packet/source census.

The relevant files are

`V98CanonicalDirectSource.lean`	canonical source and common-weight firewall
`V98CanonicalAllMRows.lean`	all-m row family
`V98DirectEnergyPin.lean`	exact energy objects and pinning interface
`V98BPPProvenance.lean`	external BPP participation provenance
`V98DirectClosure.lean`	deterministic normalized-to-physical compiler
`V98Gate0ScopeSplit.lean`	Gate-0/Gate-1A scope separation
`V98Status.lean`	axiom/status audit

12.2 What Lean does not prove

The formal development does *not* manufacture an inhabitant of the deep analytic BPP source input. In particular, the trigonometric Bernstein estimate, the source-specific smooth truncation, and Huxley's short-interval prime theorem are not currently formalized in the project. They are recorded with external/source-specific provenance, and no global axiom is introduced. The state-diagonal issue is not external: the finite moving-family bank splits diagonal and off-diagonal exactly, and the corrected projective identity places the diagonal in the $\Delta = 0$ branch.

Likewise, the formal "energy pin" is an explicit equality/interface tying a generic BPP energy to the canonical Gate-1A energy; the development includes guards showing that an arbitrary abstract bound cannot manufacture this pin. Thus a successful compiler is not confused with an independently proved analytic estimate.

12.3 Research status versus formal status

Accordingly the appropriate status language is:

$$\boxed{\text{source-explicit conditional theorem under the displayed analytic/source pins}}$$

for the mathematics of this note, and

$$\boxed{\text{external-uninhabited in Lean}}$$

for the fully formal end-to-end analytic proof. The latter is a provenance boundary, not an exponent deficit or an additional arithmetic obstruction.

12.4 Gate-1B v8.5 H7 short-short bank

The append-only v8.5 bank separates the H7 short-short region from the high-prime complement. It machine-checks the finite common-sequence and joint-prime compiler and records the exact capacity conversion

$$Y^{-1/2} = X^{-1/18}.$$

Its closure theorem is explicitly conditional on the source/common-sequence, source-energy and multiplicative-large-sieve inputs; no unconditional `H7_CLOSED` theorem is exported. The same bank leaves the high-prime complement open and records H8 only as an uninhabited robustness interface. This formal provenance is why the stronger H8 statements elsewhere in the draft are labelled research-capacity audits rather than banked closure theorems.

12.5 Gate-1B V10 zero/source/reassembly bank

The latest append-only Gate-1B V10 bank adds an exact finite/algebraic zero-mode and reassembly layer without altering the earlier Gate-1A bank. In particular, it reproves the canonical additive-character baseline from the standard character system, proves that the canonical discrepancy has no additive zero-frequency, and proves the historical/canonical residual identity

$$S_{\text{hist}} = S_{\text{can}} - R_E.$$

A perturbation counterguard shows that changing an abstract historical comparison $E(q)$ changes R_E while leaving the nonzero-frequency packet unchanged. Thus the canonical zero mode is closed at the finite/algebraic level, whereas identifying a historical $E(q)$ with a canonical comparison is an independent source question.

The same V10 bank proves literal finite packet partitions and deterministic packet-budget/reassembly compilers, including conditional implications from exact leaf inputs to the project's local Gate-1B and Type-II predicates. It does *not* inhabit the analytic leaves. A repository-wide audit in that run also found no literal declaration named `FullFMTypeII_OneSixth`; the formal compiler consequently targets the project's existing local Type-II predicate and must not be described as a formal Ford–Maynard one-sixth theorem. The V10 build is sorry-free and introduces no user axioms.

13 Relation to Kloosterman-fraction technology

Bettin–Chandee prove non-trivial bounds for separated trilinear Kloosterman fractions with arbitrary coefficient sequences [BC18]. Wright improves this architecture when a factor of the denominator is fixed [Wri26]. Blomer–Pascadi obtain a $c^{-1/32}$ saving in the square-root critical range for fixed-modulus bilinear Kloosterman forms [BP26], while Pascadi's non-abelian amplification gives, for example, a $c^{-1/12}$ saving for balanced products of two comparable primes [Pas25]. These are important fixed-multiplier/fixed-modulus capacity inputs. They do not, by their published statements, estimate the coherent moving source-multiplier family isolated in the Gate-1B appendix. The final Gate-1A direct proof likewise does not black-box any of these Kloosterman theorems; its decisive resource is smooth participation in the moving prime family.

14 What remains outside this theorem

The following are not conclusions of the main theorem:

1. exhaustive routing of every global high-complexity source packet into the canonical direct operator;
2. closure of the separate Gate-1B analytic frontier, whose present high-conductor residual is recorded in Sections C and I;
3. Full Type-II reassembly;
4. a Hardy–Littlewood prime-pair asymptotic or twin-prime infinitude.

The point of the second item is structural. The conditional Gate-1A theorem established here supplies one all-m analytic engine once its displayed pins are verified. The complementary programme has progressed beyond the earlier single H7 natural-scale residual, but the high-conductor audit now isolates an unavoidable near-primitive full-conductor signed packet. Its five centred defect factors and conductor Möbius sign remain linear, so it is not supplied by Gate 1A, by the common-conductor square norm, or by a coefficient-blind fixed-multiplier Kloosterman estimate.

15 Conclusion

The main conceptual point is that local sparsity alone does not give a moving-family saving. After source-preserving transport, the row coefficient remains r-dependent through a uniformly smooth envelope. Band-limited prime participation supplies the missing family coherence: smoothness creates a long plateau, prime distribution fills it with many moving rows, and finite Cauchy converts participation into the required $R^{-1/2+o(1)}$ energy contraction. After one and only one outer square root, the resulting $R^{-1/4+o(1)}$ gain closes the canonical all-m common-weight Gate-1A direct estimate throughout the parameter polytope.

A substantive feature of this mechanism is that its resource is the moving prime row r, not a prime factor of m. No largest-prime-factor quantity of m enters Lemmas 6.1 and 6.3 and Proposition 6.5. Thus the argument applies uniformly to the canonical all-m row family, including smooth values of m; clean-P_3 is a positive-restriction corollary rather than the source of the saving.

On the complementary side, the latest Gate-1B audit is narrower and more source-explicit than the earlier "full-covariance pass" description. The same-q cross-character proper smooth sector and the coprime cross-q no-wrap operator remain positive children. Above the primitive-conductor threshold, writing $q = cr$ recovers a short-complement sign and an exact common-conductor square norm. The subsequent square-complement audit, however, shows that this is not the full parity-breaking parent: the covariance squares away the conductor sign $\mu(c)$, the physical inverse variable in the complete Kloosterman representation is r^2B rather than r^2, and both legal Pascadi-3.8 dictionaries are fixed-power nonclosing. The common-conductor Pascadi closure lane is therefore retired. The unavoidable first analytic child is now the near-primitive $r = 1$ full-conductor five-defect signed packet (64)–(65); its top Kloosterman term admits an exact source adapter back to the native signed determinant shell, but no required estimate is yet proved. The actual switched c_9 dictionary and the literal shifted pre-enlargement TT^* source remain separate source-definition obligations.

Downstream, Ford–Maynard's universal Type-II hypothesis remains the published sufficient wrapper. Their proof of Proposition 7.22 identifies equation (7.23) as a narrower proof-specific landing point, while the latest source archaeology suggests that the remaining Gate-to-Ford problem is first a source-exact packet dictionary rather than another abstract rank-reduction theorem. We label that interface `SOURCE-ADAPTER45`. Thus the uncertainty has narrowed substantially, but Gate 1B, the downstream source dictionary, the Ford–Maynard endgame, and twin-prime infinitude all remain open/conditional as stated.

A Formal theorem map and scope ledger

Component	Status	Role in this note
Canonical source and common W_D	source-pinned; finite identities checked	fixes the source class
All-m row family	Lean-constructed	removes clean-P_3 from the statement
PB smooth envelope, retained-state disposal and $\mathcal{T}_{\text{abs}}$ ceiling	**conditional source/analytic pin**	supplies the absolute comparator
Finite BPP compiler	Lean-proved	participation $\Rightarrow$ family energy
Huxley + fixed-state exclusion	published theorem + source pin	supplies prime participation
Outer-root energy pin	**conditional source pin**; square-root algebra proved	identifies the abstract energy with the Gate energy
Ratio-class projective algebra	Lean-proved finite algebra	includes the diagonal in $\Delta = 0$
Physical projective routing and source ceiling	**conditional source pin**	bounds the zero-curvature branch
U^{-2} exponent budget	Lean-proved arithmetic	checks the recombination margin
Literal recombination estimate	**conditional source pin**	controls the source error
Gate-1B canonical zero mode	V10 Lean-proved finite algebra	zero additive frequency removed canonically
H7 short-short joint-prime compiler	conditional analytic compiler; $X^{-1/18}$ capacity	source/common-sequence pins remain
H8 analogue	research capacity audit; formal robustness interface open	not a banked closure theorem
Same-q QK56 cross-character Gram	**internal analytic pass on proper smooth sector**	programme claim, not published/Lean-closed
Cross-q source spread	**internal finite/source pass**	programme source-spread audit; final transcription pending
Shift TT^* literal source pin	**source open**	recover the literal cyclic source before the analytic child
Shifted source-linked character child	**first major analytic open**	balanced shifted-quotient source-linked character moment
Arbitrary edge-dependent packet	outside theorem scope	upstream adapter obligation
Global packet census / Full Type II	open	not implied here

B Canonical source versus arbitrary edge dependence

It is useful to make the source distinction algebraically explicit. The canonical coefficient has the form

$$\omega_e(p, q, h) = R_e(p, q, h)\, \widehat{W}_D(h/(pq)),$$

where R_e is a deterministic scalar generated by the row and by exact reciprocity. Hence the entire family is determined once the one physical weight W_D is fixed. This is fundamentally different from prescribing an arbitrary collection

$$\{W_{D,e} : e \in \mathcal{E}^\sharp\}$$

with no relation between different rows. The latter can have effective rank comparable with the number of rows and therefore cannot be reduced to the canonical theorem by functional analysis alone. This is why the global source compiler must either identify a packet with the common-weight source, reduce it to finitely many smooth templates, or route it elsewhere.

C Complementary programme boundary: Gate 1B

This appendix is a research-status calculation, not an ingredient in Theorem 10.1. "Exact" below refers to the displayed squarefree/unit source dictionary and Fourier convention. The natural-scale estimate in Proposition C.5 additionally uses the explicit source norm in Hypothesis C.4.

C.1 Low/medium conductor and the high-conductor threshold

Write $q = ce$, $c \asymp C$, $q \asymp Q$. After the corrected state count,

$$\sum_{c,\chi^*,e} |\beta(ce)|^2 \ll QC\, X^{o(1)}.$$

The coefficient-blind estimate has the relative shape

$$\frac{|\mathcal{H}_s(C)|}{Y^9} \ll \begin{cases} C/Q, & C \le Y^4, \\ C^2/(QY^4), & C \ge Y^4. \end{cases}$$

The formulae cross at $C = Y^4$. The large-C estimate loses all saving at

$$C_0 = Q^{1/2}Y^2,$$

so C_0 is the zero-saving/high-conductor threshold, not the crossing point.

C.2 Exact source split and induced-Gauss cancellation

The squarefree coefficient is

$$\beta(q) = \sum_{dp=q} \mu(d) \log p,$$

with $p > V$ and $q/p > U$. In high conductor, $e = q/c \le Y^2$, whereas $V = Y^{5/2-o(1)}$, hence $p > V > e$ and the lane $p \mid e$ is empty. Thus

$$c = pc_0, \qquad d = c_0 e,$$

and

$$\beta(ce) = \mu(e)\rho_{C,e}(c), \qquad \rho_{C,e}(c) = \sum_{pc_0=c}^{\text{physical}} \mu(c_0) \log p. \tag{21}$$

For a primitive χ^* (mod c) induced to ce,

$$\tau_{ce}(\chi) = \mu(e)\chi^*(e)\tau_c(\chi^*). \tag{22}$$

Consequently the short-cofactor sign is spent exactly:

$$\beta(ce)\tau_{ce}(\chi) = \rho_{C,e}(c)\chi^*(e)\tau_c(\chi^*).$$

It cannot be used a second time as cancellation.

C.3 Prime-character collapse

Write $c = pc_0$, $(p, c_0) = 1$, and factor a primitive character as $\chi = \chi_p\chi_0$. Let $N = r_1 \cdots r_7 x \asymp Y^8$. The p-local argument is

$$A_p \equiv c_0 e N\, \overline{2h} \pmod p.$$

For $A \in \mathbb{F}_p^\times$,

Lemma C.1 (Prime-character collapse)**.**

$$\sum_{\chi_p \neq \chi_{0,p}} \frac{\tau_p(\chi_p)}{\sqrt{p}}\chi_p(A) = \frac{p-1}{\sqrt{p}}e_p(\bar{A}) + \frac{1}{\sqrt{p}}.$$

Proof. Summing first over all multiplicative characters gives $(p-1)e_p(\bar{A})$. The principal character has Gauss sum -1, so removing it adds 1. □

The main term therefore becomes the linear additive phase

$$e_p\big(2h\, \overline{c_0 e N}\big)\, .$$

The correction is smaller than the main local multiplier by p^{-1}; since $p > V = Y^{5/2-o(1)}$, it is power-negligible under the same absolute source ceiling used below.

C.4 Hybrid Poisson formula and primitive projector

Use $\widehat{v}(\xi) = \int v(x)e(-x\xi)\, dx$.

Lemma C.2 (Hybrid Poisson)**.** *Let $p \nmid c_0$, and let χ_0 be primitive modulo c_0. With the consistent Fourier sign convention,*

$$\sum_{h\in\mathbb{Z}} v(h/H)\overline{\chi_0(h)}e_p(ah) = \frac{H}{c_0}\tau_{c_0}(\overline{\chi_0}) \sum_{\substack{m\in\mathbb{Z}\\ m\equiv ac_0\ (p)}} \chi_0(-m\bar{p})\widehat{v}\left(\frac{mH}{pc_0}\right).$$

Proof. Apply Poisson summation in residue classes modulo pc_0. The finite Fourier sum factors by CRT. The p-factor enforces the displayed congruence, while the c_0-factor is the twisted Gauss sum $\tau_{c_0}(\overline{\chi_0})\chi_0(-m\bar{p})$. The factor p from the p-sum cancels the p in the Poisson denominator. □

Here $a = 2\overline{c_0 e N}$ (mod p), so the surviving frequency obeys

$$meN \equiv 2 \pmod p, \qquad |m| \ll (Y/e)X^{o(1)} < p.$$

For odd squarefree c_0 and $(A, c_0) = 1$,

Lemma C.3 (Primitive-character projector)**.**

$$\sum_{\chi_0 \pmod{c_0}}^{*} \chi_0(A) = \sum_{\substack{d|c_0 \\ d|A-1}} \mu(c_0/d)\phi(d).$$

Consequently

$$\frac{\mu(c_0)}{c_0} \sum_{\chi_0}^{*} \chi_0(A) = \frac{1}{c_0} \sum_{\substack{d|c_0 \\ d|A-1}} \mu(d)\phi(d).$$

Proof. This is Möbius inversion on the conductor of a character. The second identity uses squarefreeness: $\mu(c_0)\mu(c_0/d) = \mu(d)$. ☐

With $A = meN/2$, the two projectors combine into $pd \mid meN - 2$.

C.5 Dual determinant and the fixed-power repair

The main term is reduced to

$$\mathcal{D}_7(C) = \sum_{\substack{pc_0 \asymp C \\ p>V \\ e \asymp Q/C}} \log p \left(1 - \frac{1}{p}\right) \sum_{N \asymp Y^8} a_7(N) \sum_{\substack{|m| \ll (Y/e)X^{o(1)} \\ p|meN-2}} \widehat{v}(me/Y)$$
$$\times \frac{1}{c_0} \sum_{\substack{d|c_0 \\ d|meN-2}} \mu(d)\phi(d). \tag{23}$$

Hypothesis C.4 (H7 source norm and cell multiplicity)**.** *The literal seven-defect plus one-model coefficient satisfies*

$$\sum_{N \asymp Y^8} |a_7(N)| \ll Y^8 X^{o(1)},$$

$\widehat{v}$ *is rapidly decreasing with bounded supremum, and the omitted physical dyadic/coprimality selectors have* $X^{o(1)}$ *total multiplicity. The primitive character family in Lemma C.3 is the literal family produced by the source dictionary (with any parity restriction separately accounted for).*

Proposition C.5 (Natural-scale fixed-power repair)**.** *Under Hypothesis C.4,*

$$|\mathcal{D}_7(C)| \ll Y^9 X^{o(1)}.$$

Proof. For fixed (e, m, N), the nonzero integer $meN - 2$ has $O(1)$ prime divisors $p > V = Y^{5/2-o(1)}$. After p is fixed, write $c_0 = ds$. The absolute divisor projector is bounded by

$$\sum_{d|meN-2} \sum_{s \asymp C/(pd)} \frac{\phi(d)}{ds} \ll X^{o(1)},$$

because the s-sum is harmonic on a dyadic interval and the divisor count is $X^{o(1)}$. Summing $e \asymp E$ and $|m| \ll Y/e$ costs $O(YX^{o(1)})$. Finally use the ℓ^1 source norm of a_7. ☐

Thus the former coefficient-blind fixed-power loss Q/Y^4 is recovered, but the target is $Y^9(\log X)^{-A}$. The remaining problem is therefore the arbitrary-log estimate

$$\boxed{\mathcal{D}_7(C) \ll_A Y^9(\log X)^{-A}.}$$

Equivalently, one may formulate a centred seven-defect determinant variance for

$$nN - pd\ell' = 2, \qquad n \ll Y, \qquad N \asymp Y^8.$$

This estimate is open. The transform also has an anti-loop: the full divisor term $d = c_0$ reconstructs the original two-model determinant skeleton, so the Poisson/character transform is not an iteration-to-closure by itself.

C.6 Historical H7 natural-scale status and the later promotion

The natural-scale proposition above records an important historical reduction: it removes the former fixed-power high-conductor deficit. It is no longer the controlling final status for the entire H7 analysis. In the short-short region

$$\alpha < \frac{4}{9}, \qquad \beta < \frac{4}{9},$$

the v8.5 bank isolates a joint-prime character packet and proves a deterministic multiplicative-large-sieve compiler. Under the explicit common-sequence, source-energy and multiplicative-large-sieve inputs, the resulting target has relative capacity

$$Y^{-1/2} = X^{-1/18}.$$

This is a *conditional analytic compiler*, not an unconditional H7 closure theorem. The high-prime complement is disjoint from this region and remains a separate source/routing problem in the formal v8.5 bank.

The latest research audit further indicates that the H8 analogue has the same fixed-power capacity after the corresponding source-energy audit. We record that as research-level analytic capacity only: the v8.5 formal development still leaves the H8 robustness interface uninhabited. Thus the publication status is deliberately asymmetric:

$$\begin{aligned} \text{H7 short-short:} &\quad \text{conditional compiler with } X^{-1/18} \text{ capacity,} \\ \text{H8:} &\quad \text{research capacity audit; formal source/robustness pin open.} \end{aligned}$$

C.7 The full $4/9$ complement and the square-root transition

The late source census organises the complement by prime exponents. At the programme level one separates a high-d wing and a high-p wing. Joint-prime machinery has favourable capacity on the high-d side and on the high-p side away from the square-root transition. The strip in which the large prime approaches $X^{1/2}$ is not absorbed by the short-short H7 compiler and is instead placed in the shifted-quotient parent introduced below.

This subsection is a routing statement, not a completed analytic theorem. In particular, the v8.5 formal bank still records the high-prime complement as open. The purpose of the parent formulation is to avoid pretending that a bound on one exponent cell automatically transports to a disjoint cell.

C.8 The Full-Nine shifted-quotient parent

The high-prime switch and the pure H9 specialization are most economically viewed inside one shifted-quotient parent of the schematic form

$$\mathcal{S}_{\mathrm{SQ}} = \sum_{mp \asymp X} \gamma(m)\lambda(p) \log p \; c_{\mathrm{res}}(mp-2). \tag{24}$$

Here γ and λ retain the actual source-generated convolution and sign structure. The high-prime M-switch is a subrange of (24), while H9 is a pure-nine-defect specialization. Type-I and joint-large-sieve mechanisms are plausible on the strongly unbalanced sides; the balanced square-root strip is the hard transition.

Equation (24) is a programme parent operator, not a theorem asserted closed in this manuscript.

C.9 The QK56 full-covariance parent

A second parent keeps the complete covariance of the one-copy QK output. If $\mathcal{Q}(q;h,k)$ denotes the one-copy output after the exact local transforms, the parent is schematically

$$\mathcal{C}_{\mathrm{QK}} = \sum_{q_1,q_2} \sum_{h_1,k_1,h_2,k_2} A(q_1,q_2;h_1,k_1,h_2,k_2)\, \mathcal{Q}(q_1;h_1,k_1)\overline{\mathcal{Q}(q_2;h_2,k_2)}. \tag{25}$$

The decomposition

$$q_1 = q_2 \quad \text{versus} \quad q_1 \neq q_2$$

is precisely the same-q diagonal versus cross-modulus off-diagonal of this single covariance parent. The same-q object does not collapse to a scalar residue-energy theorem in the source analysis; the surviving object is a multi-factor character Gram. The cross-modulus term retains the moving source data and likewise cannot be replaced by an arbitrary fixed-modulus coefficient sequence.

C.10 Pair-modulus capacity and the moving source-multiplier frontier

Both the balanced shifted-quotient TT^* child and the QK cross-modulus child lead to product moduli. For a *fixed* multiplier, current published Kloosterman technology has genuine power capacity: Blomer–Pascadi obtain a $c^{-1/32}$ saving in the square-root critical range [BP26], and Pascadi's non-abelian amplification gives, for balanced products of two comparable primes, a $c^{-1/12}$ saving [Pas25]. These results justify a backend-capacity statement only.

The actual parent carries a coherent source family

$$\sum_{\Theta} A_c(\Theta)\, \mathcal{K}_{c,\Theta}, \tag{26}$$

where the multiplier $A_c(\Theta)$ varies with the physical packet. A theorem for each fixed Θ does not by itself bound (26). This pair-modulus formulation remains a useful historical umbrella. The later audit recorded in Section F splits the QK56 covariance into same-q and cross-q children and moves the principal analytic candidate to a source-linked shifted child. Accordingly the labels

`SOURCE-QUADRATIC-MULTIPLIER45`, `FM-PERRON-PAIRMOD-SOURCE-MULT45`

are retained only as umbrella/source-grammar labels, not as the controlling first-open theorem of Draft 10.

C.11 Updated Gate-1B ledger

The current publication-facing ledger is:

Component	Current status
Orders 1–4 / low-order finite geometry	banked structural material
H6 regroup geometry	banked structural material
H7 short-short	conditional joint-prime compiler; $X^{-1/18}$ capacity
H8 short-short analogue	research capacity audit; formal robustness/source pin open
High-d/high-p complement	routing/capacity programme; square-root strip open
Full-Nine shifted-quotient parent	research parent; source-linked shifted child remains open
QK56 high-conductor covariance	**OPEN; common-conductor block is secondary after near-primitive audit**
Canonical additive zero mode	V10 exact finite/Lean pass
Historical $E(q)$ residual	source-identification fork if historical formulation retained
Same-q QK56 cross-character proper-smooth child	**internal analytic pass; not a standalone high-conductor sign closure**
Coprime cross-q no-wrap operator	**internal operator pass; high-conductor signed source family still open**
High-conductor signed parent	**RECOVERED internally**
Common-conductor signed covariance	exact square norm; **secondary** $r > 1$ **child, not full parity parent**
Old SHORT-RJ pre-Cauchy dictionary	**FALSE / RETRACTED**
Square-complement Pascadi route	**retired: naive dictionary false; legal lanes power-nonclosing**
Shift TT^* literal source pin	**SOURCE OPEN**
Near-primitive full-c five-defect signed child	**FIRST HIGH-CONDUCTOR ANALYTIC OPEN**
Shifted source-linked character child	source/analytic open, downstream of literal source pin
S1/S2 literal source pins	narrow source interfaces remain
Packet census / Gate-1B reassembly	open
Gate 1B	**NOT CLOSED**

The older D12 bulk/spike/Pascadi shortcut remains retired; this does not erase the underlying cross-modulus contribution, which is now represented in the covariance/pair-modulus parent.

D Gate 0, Gate 2, and the Ford–Maynard endgame

The preceding sections isolate the two Type-II analytic engines. To make the programme as close to a complete conditional theorem as the present evidence permits, we now state the remaining Gate-0/Gate-2 interface explicitly. This section is downstream bookkeeping and source analysis; it is not used in the proof of the conditional Gate-1A theorem.

D.1 Gate 0: deterministic Type-I compiler and research status

The separate Gate-0 bank works with a shifted-prime sequence a_n, a simpler comparison sequence b_n, and $w_n = a_n - b_n$. Its deterministic compiler has the following logical form. A suitable residue-class weighted maximal Bombieri–Vinogradov input for a_n, together with the comparison progression mean for b_n and the exact reindexing bridge, implies the Ford–Maynard Type-I predicate for w_n. The finite compiler is machine-checked, while the deep analytic inputs remain external to Lean.

Accordingly we record the Gate-0 status as

$$\boxed{\text{research-level analytic pass; external-uninhabited in Lean.}}$$

This status does not contain any Type-II information and therefore cannot by itself activate the Ford–Maynard lower-bound theorem.

D.2 The operating Ford–Maynard parameters

For the downstream application we choose the central operating width

$$\nu_0 = \frac{1}{6}.$$

For $\varepsilon > 0$ set

$$P_\varepsilon = \left(\frac{1}{2} - \varepsilon,\, \varepsilon,\, \frac{1}{6} - 2\varepsilon\right). \tag{27}$$

The corresponding Type-II interval in the Ford–Maynard convention is

$$[\theta, \theta + \nu] = \left[\varepsilon, \frac{1}{6} - \varepsilon\right]. \tag{28}$$

The published central threshold in Ford–Maynard Theorem 2.7 is 0.1663; our operating value is safely above it, irrespective of the harmless strict/non-strict convention at the printed endpoint:

$$\frac{1}{6} - 0.1663 = \frac{11}{30000} > 0. \tag{29}$$

Ford–Maynard also use precisely the shrunk family $(1/2 - \varepsilon, \varepsilon, \nu - 2\varepsilon)$ in Theorem 2.7 [FM24]. Thus the appearance of the third coordinate $\nu - 2\varepsilon$ is part of the published parameter geometry rather than a new sieve hypothesis introduced here.

D.3 A finite roughness consequence

Write

$$\sigma = \nu_0 - 2\varepsilon.$$

For $0 \le \varepsilon \le \nu_0/100$, the Gate-0/2 finite bank proves

$$\sigma \ge \frac{49}{300}. \tag{30}$$

The following elementary lemma is fully independent of the deeper Ford–Maynard analysis.

Lemma D.1 (Rough numbers have at most six prime factors)**.** *Let $n > 1$ and suppose every prime divisor of n is at least n^σ, where $\sigma \ge 49/300$. Then*

$$\Omega(n) \le 6,$$

where prime factors are counted with multiplicity.

Proof. If $k = \Omega(n)$, then

$$n = \prod_{j=1}^{k} p_j \geq n^{k\sigma}.$$

Hence $k\sigma \leq 1$, so

$$k \leq \frac{1}{\sigma} \leq \frac{300}{49} < 7.$$

Since k is an integer, $k \leq 6$. □

This finite statement is machine-checked in the Gate-0/2 bank. It is useful because the final rough N_2 contribution lies in a bounded-complexity class once the required Ford–Maynard roughness identity has been supplied.

D.4 The rough-weight input and the N_2 cell sum

The remaining source-specific rough-weight input has the form

$$H(n) = (1 * g)(\mathbf{v}(n))\, \mathbf{1}_{\{P^-(n) \geq n^\sigma\}}, \qquad |H(n)| \ll_{g,\nu_0} 1. \tag{31}$$

Ford–Maynard's Section 7 contains the corresponding rough-support mechanism; Lemma 7.18 is used in the proof of Theorem 7.3 to restrict the support by a least-prime-factor condition, while Theorem 8.2 combines the Section 7 machinery with a small exceptional region [FM24]. In the formal bank, however, the exact shifted-prime instantiation (31) is deliberately left uninhabited until its complete source dictionary is certified.

The Gate-0/2 development separately records the aggregate N_2 cell-sum repair: after choosing the largest prime as the final sieve variable, using a coarse geometric mesh and aggregating the cell errors, the desired N_2 upper bound is reduced to the two-linear-forms upper-sieve and prefix-volume/Mertens–PNT inputs. The finite compiler and the ε-uniformity bookkeeping are banked; the analytic source input is not promoted to a Lean theorem.

D.5 Gate 2 as a conditional endgame: published and proof-specific routes

The published Ford–Maynard framework is formulated with a general Type-II hypothesis for arbitrary divisor-bounded coefficient sequences. We keep that statement as the clean published sufficient wrapper and denote the required operating-width input schematically by

$$\mathrm{FullFMTypeII}_{1/6}.$$

Together with the source-exact rough N_2 input and the frozen Gate-0 comparison/Type-I inputs, this gives the programme-level conditional implication

$$\begin{aligned} &\mathrm{FullFMTypeII}_{1/6} + \mathrm{FMRoughN2} + \text{Gate-0 comparison inputs} \\ &\qquad\qquad \Longrightarrow \text{positive shifted-prime/twin mass at scale } x. \end{aligned} \tag{32}$$

This is the published-route conditional wrapper; the analytic antecedents are not proved by the present manuscript.

A proof audit suggests a potentially weaker route. In the proof of Ford–Maynard Proposition 7.22, after the finite factor decomposition the left-hand side of their equation (7.23) is written as a grouped bilinear sum

$$\sum_{n = n'n'' \asymp x} Y_1(n')Y_2(n'')w_{n'n''},$$

and the published Type-II bound is then invoked on this grouped object. Thus a direct theorem controlling the complete family of structured sums actually created by the proof would be sufficient for that invocation. We denote the corresponding *proof-specific candidate interface* by

$$\boxed{\texttt{FM-SIEVEGEN-TYPEII45}}.$$

This is a weaker-hypothesis reanalysis of the proof, not a Ford–Maynard theorem statement and not yet a completed theorem of this manuscript.

A further proposed narrowing, labelled

$$\boxed{\texttt{FM-PERRON-GENERATED-TYPEII45}},$$

tracks the pre-supremum coefficients to a generated grammar consisting of Möbius factors, box/support cutoffs, Mellin/Perron twists, least/greatest-prime-factor phases, perfect-power pullbacks and bounded-depth convolutions. This generated grammar is a research target pending a complete grammar/source certificate; it is not asserted proved.

Accordingly, a second conditional route may be written only after introducing an explicit *replacement certificate* $\mathfrak{C}_{\mathrm{FM}}$ which verifies that every Type-II use in the rerun of the relevant Ford–Maynard lower-bound proof is covered by the structured theorem. Conditional on such a certificate,

$$\begin{aligned}\mathfrak{C}_{\mathrm{FM}} + \mathrm{FMRoughN2} + \text{Gate-0 comparison inputs}\\ \Longrightarrow \text{positive shifted-prime/twin mass at scale } x.\end{aligned} \tag{33}$$

Neither $\mathfrak{C}_{\mathrm{FM}}$ nor the Perron-generated theorem is constructed in the current paper.

Theorem D.2 (Conditional endgame, publication-safe form)**.** *Assume the frozen Gate-0 comparison/Type-I inputs and the source-exact rough N_2 input. If either*

(a) the published universal Ford–Maynard Type-II hypothesis holds on the interval $[\varepsilon, 1/6-\varepsilon]$, or

(b) a separately certified proof-specific replacement package $\mathfrak{C}_{\mathrm{FM}}$ covers every Type-II invocation required in the rerun of the downstream proof,

then the corresponding conditional Gate-2 compiler yields positive shifted-prime/twin mass at scale x. If the chosen package holds at arbitrarily large scales, the finite positivity implication yields infinitely many distinct twin-prime pairs.

Proof. This is a conditional logical statement. In route (a), the published Ford–Maynard Type-II antecedent supplies the required bilinear estimates. In route (b), the replacement certificate is, by definition, required to verify that the structured theorem covers every Type-II invocation used by the rerun. The rough-N_2 and comparison inputs supply the remaining analytic antecedents. The finite Gate-2 compiler then produces positive mass at the scale. Repeating on arbitrarily large scales yields distinct finite-support witnesses. The present manuscript proves neither route's missing analytic antecedents. □

D.6 Where Gate 1A and Gate 1B land in the downstream Type-II input

The published universal Type-II wrapper remains sufficient and should not be removed from the dependency graph. What has changed is the programme's candidate minimal interface. Gate 1A supplies a conditional all-m analytic engine; the late Gate-1B architecture aims to supply the complementary source-coherent pair-modulus multiplier estimates. A source-exhaustive compiler must then show that the actual generated coefficients required by the Ford–Maynard rerun lie in the chosen structured class.

The honest dependency graph therefore has two possible Type-II entrances:

$$\begin{array}{ccc} \text{Gate 0 inputs} & \text{Gate 1A engine} & \text{Gate 1B pair-modulus frontier} \\ & \searrow \quad \downarrow \quad \swarrow & \end{array}$$

$$\begin{cases} \mathrm{FullFMTypeII}_{1/6}, & \text{published sufficient wrapper,} \\ \mathfrak{C}_{\mathrm{FM}}, & \text{proof-specific replacement, if certified,} \end{cases}$$

$$\downarrow$$

$$\text{Gate 2 Ford–Maynard endgame}$$

$$\downarrow$$

$$\text{positive twin mass at a scale.}$$

At present, the moving source-multiplier estimate, the generated-grammar certificate, the literal source pins and source-exhaustive reassembly remain open. Thus neither entrance is presently activated by the manuscript.

D.7 Publication-facing Gate-0/2 status

For clarity, the downstream status is summarized as follows:

Object	Current status
Gate-0 deterministic Type-I compiler	Lean-banked finite compiler
Gate-0 deep analytic inputs	research pass / external-uninhabited
$1/6 > 0.1663$ and margin $11/30000$	Lean-banked arithmetic
$\sigma \geq 49/300$	Lean-banked arithmetic
$\Omega(n) \leq 6$ for σ-rough n	Lean-banked theorem
Ford–Maynard rough-weight source dictionary	external/source pin; not a Lean inhabitant
N_2 cell-sum and ε-uniform splice	conditional research pass; Lean-uninhabited
Full FM Type II at width $1/6$	OPEN / uninhabited
Gate 2	CLOSED ONLY CONDITIONALLY on the displayed antecedents
Full Type-II reassembly	OPEN
Twin-prime infinitude	NOT PROVED

Status note D.3 (Full-programme closure status)**.** The role of this section is to make the strongest currently defensible conditional closure explicit. It does not upgrade any external, source-inspected or programme-generated interface to a theorem. Gate 1B is not closed. A subsequent source audit refines the earlier pair-modulus frontier into a same-q cross-character Gram, a cross-q source-spread certificate, and a source-linked shifted-quotient child; none is promoted to a theorem here. Downstream, universal Ford–Maynard Type II remains a published sufficient wrapper, while `FM-SIEVEGEN-TYPEII45` and `FM-PERRON-GENERATED-TYPEII45` are progressively narrower proof-specific research targets. The literal Gate-output/Ford-source adapter, the generated grammar certificate, source pins and source-exhaustive reassembly are not yet certified, and twin-prime infinitude is not proved.

E Late update: refined Gate-1B frontier and Ford–Maynard landing strip

This late update records source audits completed after the main body of the draft. It refines the dependency graph but does not alter the statement or proof status of the conditional

Gate-1A theorem. In particular, none of the programme labels introduced below is promoted to a theorem merely by being recorded here.

E.1 Refined Gate-1B source split

The earlier pair-modulus formulation remains a useful umbrella, but a later source audit suggests a finer decomposition.

For the QK56 covariance parent, the cross-modulus branch $q_1 \neq q_2$ lies in a product-modulus regime in which the relevant multiplicative product ranges are shorter than the product modulus. In that no-wrap regime, weighted product-energy arguments reduce the remaining issue to a source-level spread/multiplicity statement for the push-forward multiplier. We record the temporary programme label

$$\boxed{\texttt{CROSSQ-THETA-SPREAD45}}.$$

The subsequent internal audit recorded in Section F marks this source-spread step as an internal finite/source pass, subject to final literal transcription. It remains neither a published theorem nor a Lean-closed analytic result.

The same-modulus branch $q_1 = q_2 = q$, by contrast, retains genuine modular wraparound. After character expansion the surviving object has schematic shape

$$\sum_{q \asymp Q} |\beta(q)|^2 \sum_{\chi_1,\chi_2 \bmod q} \widehat{R}_q(\chi_1)\,\overline{\widehat{R}_q(\chi_2)}\,\mathcal{W}_q(\chi_1,\chi_2).$$

The diagonal $\chi_1 = \chi_2$ does not exhaust this Gram, and fixed-multiplier Kloosterman estimates do not remove the cross-character terms. The subsequent internal diagonal/off-diagonal audit recorded in Section F gives this child an internal analytic pass, with the literal QK56/S1/S2 source dictionary still load-bearing:

$$\boxed{\texttt{SAMEQ-QK56-CROSSCHAR-GRAM45:\ \ internal analytic pass}}.$$

This remains a programme claim subject to source transcription and hostile audit, not an externally published theorem.

The balanced shifted-quotient branch likewise carries source-linked four-cycle multipliers rather than four independent arbitrary multipliers. Its first required action is now the literal source recovery

$$\boxed{\texttt{SHIFT-TTSTAR-LITERAL-SOURCE-PIN}},$$

after which the narrower research child

$$\boxed{\texttt{SHIFT-SOURCE-LINKED-CHAR45}}$$

is the first major analytic candidate. Published fixed-modulus Kloosterman results remain backend-capacity inputs only: Blomer–Pascadi give a $c^{-1/32}$ saving in the square-root critical range [BP26], while Pascadi's non-abelian amplification gives $c^{-1/12}$ for balanced products of two comparable primes [Pas25]. Neither published statement controls the above moving source-linked family automatically.

Thus the refined Gate-1B research ledger is

`SAMEQ-QK56-CROSSCHAR-GRAM45`	internal analytic pass; source pin pending,
`CROSSQ-THETA-SPREAD45`	internal finite/source pass; transcription pending,
`SHIFT-TTSTAR-LITERAL-SOURCE-PIN`	source first action open,
`SHIFT-SOURCE-LINKED-CHAR45`	first major analytic open,
$S1/S2$ literal normalisation	narrow source pins,
degeneracy/collision/packet census	finite routing/reassembly.

This supersedes only the research-level description of the common Gate-1B frontier. Gate 1B remains open.

E.2 Published Ford–Maynard five-collar geometry

One downstream geometric source issue can be removed from the open ledger. For

$$P_\varepsilon = \left(\frac{1}{2} - \varepsilon,\, \varepsilon,\, \nu - 2\varepsilon\right),$$

Ford–Maynard's proof of Theorem 8.3 shows that every vector $x \in H_2$ has a component of one of the five forms

$$\frac{1}{2} + u, \qquad \nu + u, \qquad 2\nu + u, \qquad 1 - \nu + u, \qquad 1 - 2\nu + u, \qquad |u| \le 6\varepsilon. \tag{34}$$

This is a published source statement [FM24]. Moreover, Ford–Maynard Definition 5.1 defines

$$\mathbf{v}(n) = \left(\frac{\log p_1}{\log n}, \dots, \frac{\log p_k}{\log n}\right), \qquad n = p_1 \cdots p_k,$$

so the coordinates in (34) correspond to individual prime factors. At $\nu_0 = 1/6$ the collar centres are

$$\frac{1}{2}, \qquad \frac{1}{6}, \qquad \frac{1}{3}, \qquad \frac{5}{6}, \qquad \frac{2}{3}.$$

We therefore record the programme status

$$\boxed{\texttt{FM-H2-FIVE-COLLARS45:\ \ published-source pass}}.$$

E.3 A source-specific repair of the rough N_2 sector

The common normalization $a_n, b_n \mapsto a_n/\log x, b_n/\log x$ does not by itself place the shifted-prime comparison pair into the bounded Ford–Maynard class: it also reduces the prime mass by an additional factor of $\log x$. Accordingly, that normalization is not used as the N_2 repair.

Instead set

$$\nu_0 = \frac{1}{6}, \qquad \sigma = \nu_0 - 2\varepsilon.$$

For $0 < \varepsilon \le \nu_0/100$ one has $\sigma \ge 49/300$, and the finite bank proves that every σ-rough integer has at most six prime factors counted with multiplicity.

The squareful rough subfamily is negligible. Indeed, if $p^2 \mid n \le x$ and $p \ge x^\sigma$, then

$$\sum_{x^\sigma \le p \le x^{1/2}} \left(\frac{x}{p^2} + 1\right) \log x \ll x^{1-\sigma} \log x + x^{1/2} \log x = o\left(\frac{x}{\log x}\right). \tag{35}$$

For the squarefree sector, the five-collar geometry permits one to choose canonically a collar prime q and then the largest remaining prime r, writing $n = qdr$. The present research reduction controls the shifted-prime part by a uniform upper-bound sieve for the two linear forms

$$r, \qquad qdr + 2.$$

On the rough sector the associated local factor is uniformly bounded, while the harmonic sum over the narrow prime collar contributes $O_{\nu_0}(\varepsilon + 1/\log x)$. This yields the research-level estimate

$$\sum_{n \in N_2} a_n |H(n)| \ll (\varepsilon + o(1)) \frac{x}{\log x},$$

conditional on the standard uniform two-linear-forms upper-sieve input in the stated rough range. The comparison side is compatible with the Ford–Maynard rough-number count and the same five-collar volume estimate. We record this as

$$\boxed{\texttt{FM-N2-SHIFTED-PRIME-COLLAR-SIEVE45}}$$

with research status "pass modulo a standard uniform two-linear-forms upper sieve". This remains external/uninhabited in Lean until the literal shifted-prime source dictionary and analytic upper-sieve input are instantiated.

E.4 The proof-specific Ford–Maynard landing point

The published Ford–Maynard Type-II hypothesis remains the clean sufficient wrapper. However, Proposition 7.22 provides a more economical *proof-specific* landing point. After the finite factor decomposition, Ford–Maynard reduce the relevant estimate to their equation (7.23), namely a finite-depth product sum with one selected product constrained to the Type-II interval. They then group the factors into

$$Y_1(n') = \sum_{n'=\prod_{e\in E} n_e} \prod_{e\in E} \mathfrak{x}_e(n_e), \qquad Y_2(n'') = \sum_{n''=\prod_{e\notin E} n_e} \prod_{e\notin E} \mathfrak{x}_e(n_e),$$

and invoke the universal Type-II bound only after reaching

$$\sum_{n=n'n''\asymp x} Y_1(n')Y_2(n'')w_{n'n''}.$$

This is a direct reading of the proof of Proposition 7.22 [FM24]. Consequently, a theorem controlling *all actual generated instances* of Ford–Maynard equation (7.23) is sufficient for this specific invocation:

$$\boxed{\text{actual generated Ford–Maynard (7.23)} \Longrightarrow \text{Proposition 7.22} \Longrightarrow \text{Proposition 7.19.}}$$

This is a proof audit, not a new theorem of Ford–Maynard and not a claim that the antecedent has been proved here.

The factor depth is fixed by the sieve parameters. In the proof,

$$N = k\ell + r + s,$$

with k, ℓ, r, s bounded in terms of the fixed sieve data; thus the earlier concern that the generated landing point necessarily has unbounded convolution depth is not an independent obstruction.

We retain the temporary label

$$\boxed{\texttt{FM-PERRON-GENERATED-TYPEII45}}$$

with status:

$$\boxed{\text{proof-specific sufficient target; generated grammar/source certificate open.}}$$

E.5 The remaining Gate-output/Ford-source adapter

The present Gate-1A theorem controls a canonical common physical weight source, and the Gate-1B analysis has its own literal source classes. Ford–Maynard's factor decomposition, however, generates a finite family of one-variable factors before equation (7.23). The next downstream question is therefore first a source dictionary problem:

$$\omega_{\Omega,j} \longmapsto \begin{cases} \text{Gate-1A source,} \\ \text{Gate-1B source,} \\ \text{direct exceptional source.} \end{cases}$$

We use the programme label

$$\boxed{\texttt{SOURCE-ADAPTER45}}$$

for this literal Gate-output/Ford-source splice. Its current status is:

$$\boxed{\text{primary candidate downstream interface; source-exhaustive dictionary open.}}$$

Until the literal audit is complete, it is not known whether this is merely a finite compiler problem or whether one generated weight family exposes a stronger analytic child.

E.6 Revised downstream ledger

The late-update status is:

Ford H_2 five-collar geometry	published-source pass,
σ-rough $\Omega(n) \leq 6$	Lean-banked finite theorem,
N_2 squareful disposal	pass,
N_2 shifted-prime two-form lane	research pass modulo standard upper sieve,
Ford (7.23) proof-specific landing	proof-audit pass,
unbounded-depth grammar concern	retired,
`SOURCE-ADAPTER45`	primary downstream candidate interface,
`SAMEQ-QK56-CROSSCHAR-GRAM45`	internal analytic pass; source pins pending,
`SHIFT-SOURCE-LINKED-CHAR45`	first major Gate-1B analytic open,
Gate 1B	**OPEN**,
Full Type-II reassembly	**OPEN**,
twin-prime infinitude	**NOT PROVED**.

The revised shortest dependency graph is

$$\begin{gathered}
\text{Gate-0 Type-I/comparison inputs} + \text{Gate-1A literal output} + \text{Gate-1B literal output} \\
\downarrow \\
\boxed{\texttt{SOURCE-ADAPTER45 / source-exhaustive Ford-(7.23) compiler}} \\
\downarrow \\
\text{actual generated Ford–Maynard (7.23)} \\
\downarrow \\
\text{Proposition 7.22} \to \text{Proposition 7.19} + \text{source-specific } N_2 \text{ repair} \\
\downarrow \\
\text{Ford–Maynard lower-bound rerun at } \nu = 1/6 \\
\downarrow \\
\text{positive shifted-prime/twin mass at a scale.}
\end{gathered}$$

Every arrow after the source-exhaustive compiler remains conditional on the literal analytic antecedents displayed above. This diagram does not assert the Twin Prime Conjecture.

Status note E.1 (Late-update publication firewall)**.** The late update narrows the downstream uncertainty, but does not reduce the programme to one remaining theorem. In particular:

- Gate 1B is not closed;
- `SAMEQ-QK56-CROSSCHAR-GRAM45` is an internal analytic pass subject to literal source pins and hostile audit, not a published or Lean-banked analytic theorem;
- `SOURCE-ADAPTER45` has not yet been classified as finite-only versus genuinely analytic;
- `FM-PERRON-GENERATED-TYPEII45` is proof-specific and the generated source grammar is not yet certified;
- the Ford–Maynard endgame remains conditional;
- twin-prime infinitude is not proved.

F Late update II (historical): QK56 promotion, shifted source-linked child, and source-exact Ford adapter

This section records an intermediate late-26/27-August programme audit. It does not alter Theorem 10.1 and does not assert Gate-1B closure, Full Type-II closure, or twin-prime infinitude. The subsequent native first-TT^* derivation in Section G supersedes the stronger full-covariance status wording below: only the same-q cross-character child and the cross-q operator theorem survive as positive analytic children; the same-q diagonal source energy and cross-q physical source spread remain open.

F.1 QK56 full covariance: internal promotion

The same-modulus covariance is first split into the character diagonal and the cross-character part. Schematically,

$$\sum_{q\sim Q} |\beta(q)|^2 \sum_{\chi_1,\chi_2 \bmod q} \widehat{R}_q(\chi_1)\overline{\widehat{R}_q(\chi_2)}\mathcal{W}_q(\chi_1,\chi_2) = \mathcal{G}_{\mathrm{diag}} + \mathcal{G}_{\mathrm{off}}.$$

The internal audit places the large HK/q contribution in $\mathcal{G}_{\mathrm{diag}}$ and routes that diagonal to the existing coefficient-blind residue-energy/JQ7 bank. The off-diagonal is identified with a centred modular-hyperbola discrepancy and is estimated internally through a square-root Kloosterman/Weil discrepancy argument. The current worst signed programme margin is

$$X^{-5/36+o(1)}.$$

We therefore record

$$\boxed{\texttt{SAMEQ-QK56-CROSSCHAR-GRAM45: internal analytic pass}}$$

subject to the literal S1/S2 source pin and final hostile audit.

For $q_1 \neq q_2$, the latest no-wrap/source audit uses the fact that fixing the source parameter Θ determines $B_1 \bmod q_1$ and $B_2 \bmod q_2$. The remaining product/numerical multiplicity is internally within the required ℓ^2/ℓ^1 spread budget. We therefore record

$$\boxed{\texttt{CROSSQ-THETA-SPREAD45: historical candidate; source spread later reopened}},$$

again without promoting the statement to a published theorem or a Lean-closed analytic result.

At programme level the combined status is

$$\boxed{\begin{array}{c}\texttt{QK56-FULL-COVARIANCE45}\\ \text{historical programme promotion; superseded by Late update III}\end{array}}$$

F.2 The shifted child: source action before analysis

The balanced shifted-quotient branch should not be modelled as four independent multipliers. Its cyclic source relation has the form

$$a_j = \Theta(B_j, B_{j+1}) \qquad (j \bmod 4),$$

and the schematic character moment is

$$\sum_{B_1,\dots,B_4} W(\mathbf{B}) \sum_h W_{\mathbf{B}}(h)\, \chi\big(\mathscr{D}\big(\Theta(B_1,B_2),\Theta(B_2,B_3),\Theta(B_3,B_4),\Theta(B_4,B_1);h\big)\big)\,.$$

The four-cycle discriminant algebra is banked in the programme, but the literal pre-support-enlargement TT^* source formula has not yet been recovered. Consequently the first action is

$$\boxed{\texttt{SHIFT-TTSTAR-LITERAL-SOURCE-PIN}},$$

which must print the literal B_i ranges, the exact cyclic Θ map, the h-range and weight, prefactor and normalization, physical target, all zero/nonunit/collision/repeated-prime strata, and labelled multiplicity. Only after this transcription is the analytic target

$$\boxed{\texttt{SHIFT-SOURCE-LINKED-CHAR45}}$$

well posed. It is the current first major analytic Gate-1B candidate.

F.3 Technology cards for the shifted source-linked child

Several published results are relevant only after the literal source has been reduced to their stated geometries.

Fouvry–Shparlinski estimate weighted multiplicative-character sums of the 2×2 determinant $ad - bc$ over four interval variables, with power saving for $N \geq p^{1/8+\varepsilon}$ [FS22]. This is a serious candidate if the cyclic discriminant is reduced source-exactly to determinant geometry.

Shparlinski treats multivariate exponential and multiplicative-character sums with monomials, allowing bounded one-variable weights and both positive and negative monomial exponents [Shp13]. This is relevant if the cyclic Θ dictionary reduces to a monomial or reciprocal-monomial kernel. Bourgain–Garaev's reciprocal sumset and multilinear Kloosterman technology is a secondary candidate once an actual reciprocal-product dictionary has been manufactured [BG12].

The very general trace-function bilinear theorem of Fouvry–Kowalski–Michel–Sawin requires substantial structural hypotheses on the geometric monodromy group (for instance a simple connected identity component, or finite quasisimple monodromy in the displayed simplified theorem) [Fou+25]. A rank-one Kummer/Legendre character does not satisfy those displayed hypotheses merely by being a trace function, so that result is not cited as a black box for the present child. Blomer–Pascadi remains a fixed-Kloosterman backend card and likewise does not resolve the source-linked discriminant family by itself [BP26].

F.4 Minimal current Gate-1B ledger

The late-update Gate-1B ledger is therefore

`QK56-FULL-COVARIANCE45`	historical promotion; full source-exact closure reopened
`SHIFT-TTSTAR-LITERAL-SOURCE-PIN`	source open; first action
`SHIFT-SOURCE-LINKED-CHAR45`	first major analytic open
S1/S2 literal normalization	narrow source pins
$D = 0$/shared-prime/nonunit/collision census	finite closure/routing
fixed/switched reassembly	finite compiler

Thus Gate 1B is still not closed. The later native first-TT^* source derivation shows that QK56 itself still contains two source-exact quantitative gaps—the same-q character diagonal and cross-q CRT source spread—before the shifted-source branch and finite reassembly can be assessed exhaustively.

F.5 Ford-generated one-variable archaeology and the Gate-output source dictionary

This source question is separate from Gate-1B analysis. The latest internal Ford-side archaeology records the following programme passes:

`FM-GENERATED-ONEVAR-ARCHAEOLOGY45`	pass,
`FM-FINITE-NUCLEAR-COMPILER45`	pass,
`FM-OUTER-COEFFICIENT-ABSORPTION45`	pass,
`WEIGHT-DEPENDENCE-FINITE-COMPILER45`	pass.

These are programme-level source/compiler audits, not published theorems. The first unresolved interface is the source-exact weighted high-complexity packet dictionary. For each actual packet it must provide the operator, ranges, coefficient, multiplicity, weight law, centering, gcd/collision restrictions, original prefactor, normalized target and route.

In particular the actual Gate weight law is not to be guessed: the relevant packet must be identified as a common W_D, a finite-template $W_{D,e}$, or a genuinely edge-dependent $W_{D,e}$. We therefore retain

`FM-TO-GATE-COORDINATE-CENSUS45:  source-blocked`

rather than calling it a new analytic theorem. Once this literal source census exists, the finite coordinate/reassembly compiler is already available at programme level.

F.6 Ford–Maynard scope

Ford–Maynard's published framework remains the general sufficient wrapper for non-negative sequences satisfying its Type-I/Type-II hypotheses [FM24]. The programme's generated-one-variable and generated (7.23) routes are proof-specific weakenings under investigation; we do not attribute them to Ford–Maynard as theorem statements. In particular, the latest source/compiler audits do not change the final publication firewall:

Gate 1B open; downstream source dictionary open;
Ford–Maynard endgame conditional; twin primes not proved.

Status note F.1 (Draft 10 late-update firewall)**.** The same-q cross-character promotion remains an internal programme result; the cross-q operator theorem is likewise internal and does not provide the physical source-spread estimate. The stronger QK56 full-covariance status in this historical section is superseded by Section G. None of these statements is a kernel-checked Lean analytic theorem or an externally published Gate-1B theorem. Likewise, `SHIFT-SOURCE-LINKED-CHAR45`, the physical high-complexity packet dictionary, the FM-to-Gate coordinate census and the Ford lower-bound rerun must not be promoted without their actual proof/source objects.

G Late update III (partial child analysis): native first-TT^* Gate-1B reduction

This section supersedes the stronger QK56 full-covariance status language in Section F. It is a source-explicit programme update, not a claim of Gate-1B closure. The decisive correction is that the first TT^* covariance has three different modulus geometries. The two positive analytic children below remain valid on their stated proper-smooth sectors. Late update IV supersedes this section as the controlling high-conductor architecture: once the complement Möbius sign is restored, same-q and common-conductor cross-q can no longer be closed sequentially.

G.1 Scope and current status

This note is deliberately not a claim of Gate–1B closure. It isolates the part of the remaining analysis that can now be written in source-exact coordinates and separates it from source-definition and reassembly tasks.

The controlling internal status is:

same-q smooth cross-character discrepancy	analytic pass on the proper smooth sector,
coprime cross-q no-wrap moving-multiplier L^2	proved at the kernel/operator level,
same-q character diagonal	source-energy open,
cross-q CRT source spread	source-norm open,
Full-Nine/Ford–Maynard reassembly	source-exhaustive compiler open,
shifted four-cycle literal TT^* source	source definition open.

Published Kloosterman and determinant-character technology is useful only after the literal source dictionary is fixed. In particular, fixed-modulus bilinear Kloosterman estimates do not by themselves supply a moving source multiplier family theorem.

G.2 Native order-five source and two-slot completion

Let

$$Y = X^{1/9}, \qquad q \asymp Q = X^{\omega}, \qquad \frac{13}{18} \le \omega < \frac{8}{9}, \qquad qR \asymp X = Y^9.$$

In the critical order-five cell, write the product of five defect factors as $C_5 \asymp Y^5$ and retain four model variables $x_1, x_2, x_3, x_4 \asymp Y$. Put

$$B = C_5 x_2 x_3 \asymp Y^7.$$

The native shifted relation is

$$\boxed{B x_1 x_4 - q\ell = -2.} \tag{36}$$

This is the determinant shell generated by the $n+2$ source; it is not a replacement by a $2n \pm 1$ model.

For fixed (B, q) the two physical Poisson variables are x_1, x_4. With smooth source factors f_1, f_4, W, set

$$F_{B,q}(x_1, x_4) = f_1(x_1) f_4(x_4) W\left(\frac{B x_1 x_4 + 2}{qR}\right) \mathbf{1}_{B x_1 x_4 \equiv -2 \pmod q}.$$

Two-dimensional completion gives

$$\mathcal{T}_{B,q} = \frac{Y^2}{q^2} \sum_{h,k \in \mathbb{Z}} \Psi_{B,q}(h,k) S(h, -2k\overline{B}; q). \tag{37}$$

On the proper unit sector $(kB, q) = 1$, Kloosterman scaling yields

$$S(h, -2k\overline{B}; q) = S(hk\overline{B}, -2; q),$$

so the actual one-copy multiplier is

$$\boxed{a = hk\overline{B} \pmod q.} \tag{38}$$

The effective ranges are

$$|h|, |k| \ll q/Y,$$

and the normalized smooth kernel has uniformly bounded derivatives on a fixed compact box. Hence the smooth source can be separated into polylogarithmically controlled nuclear packets. This separation is a source-normalization device, not an independent arithmetic saving.

G.3 The first TT^* sector law

For one nuclear packet write

$$\mathcal{F}_q(h,k) = \sum_B c_{q,B}(h,k) S(hk\overline{B}, -2; q).$$

Opening the covariance produces

$$\sum_{h,k} \left| \sum_q \mathcal{F}_q(h,k) \right|^2 = \sum_{q_1,q_2,B_1,B_2,h,k} c_{q_1,B_1}(h,k) \overline{c_{q_2,B_2}(h,k)} \mathcal{K}_{12}(hk),$$

where

$$\mathcal{K}_{12}(t) = S(t\overline{B_1}, -2; q_1) \overline{S(t\overline{B_2}, -2; q_2)}.$$

There are three genuinely different arithmetic sectors.

G.3.1 Same modulus

If $q_1 = q_2 = q$, the product is most naturally diagonalized by multiplicative characters. For $(mn, q) = 1$,

$$S(m,n;q) = \frac{1}{\phi(q)} \sum_{\chi \bmod q} \tau_q(\chi)^2 \overline{\chi}(mn). \tag{39}$$

Thus the same-q covariance is a character Gram rather than a single moving Kloosterman sum. Its character diagonal and cross-character pieces must be treated separately.

G.3.2 Coprime cross-modulus

If $q_1 \neq q_2$ and $(q_1, q_2) = 1$, CRT multiplicativity gives

$$\boxed{\mathcal{K}_{12}(t) = S(t, \Theta_{12}; q_1 q_2),} \tag{40}$$

where $\Theta_{12} \pmod{q_1 q_2}$ is uniquely determined by

$$\Theta_{12} \equiv -2q_2^2 \overline{B_1} \pmod{q_1}, \qquad \Theta_{12} \equiv -2q_1^2 \overline{B_2} \pmod{q_2}.$$

This is the actual first-TT^* moving multiplier.

G.3.3 Shared divisor

When $(q_1, q_2) > 1$, the product modulus cannot be treated by Eq. (40) without first extracting the common conductor. This branch belongs to the existing GCD/shared-prime router.

Proposition G.1 (First-TT^* sector law)**.** *The native QK56 covariance separates into:*

$$\begin{aligned} \textit{same-}q &= \textit{character Gram}, \\ (q_1, q_2) = 1 &= \textit{CRT Kloosterman}, \\ (q_1, q_2) > 1 &= \textit{shared-conductor router}. \end{aligned}$$

In particular, the first covariance is not a universal four-independent- multiplier character moment.

G.4 Same-q cross-character discrepancy

The cross-character part of the same-q Gram admits a source-specific modular-hyperbola treatment.

Let $G_q = (\mathbb{Z}/q\mathbb{Z})^\times$. For a pair of separated smooth frequency weights u, v define

$$g_q(r) = \sum_{\substack{h,k \in G_q \\ hk=r}} u(h)v(k)$$

and

$$m_q = \frac{1}{\phi(q)} \left(\sum_h u(h) \right) \left(\sum_k v(k) \right).$$

Set

$$\Delta_q(u, v) = \max_{r \in G_q} |g_q(r) - m_q|.$$

G.4.1 Prime modulus

For prime q, additive completion gives the exact identity

$$g_q(r) = \frac{1}{q^2} \sum_{a,b \bmod q} \widehat{u}(a)\widehat{v}(b)S(a, br; q).$$

After removing the zero frequencies one obtains

$$g_q(r) - \frac{UV}{q-1} = \frac{1}{q^2} \sum_{\substack{a,b \bmod q \\ a,b \neq 0}} \widehat{u}(a)\widehat{v}(b)S(a, br; q) - \frac{UV}{q^2(q-1)},$$

where $U = \sum u$ and $V = \sum v$.

Define the discrete bounded-variation seminorm

$$\mathcal{B}(u) = \|u\|_\infty + \sum_x |u(x+1) - u(x)|.$$

Summation by parts gives

$$\sum_{a \neq 0} |\widehat{u}(a)| \ll q\,\mathcal{B}(u)\log(2q),$$

and similarly for v. The Weil bound then yields

$$\Delta_q(u, v) \ll q^{1/2}\mathcal{B}(u)\mathcal{B}(v)\log^2(2q) + O\left(\frac{|UV|}{q^3}\right). \tag{41}$$

G.4.2 Composite modulus

For general proper-unit q, the modular inverse graph has Fourier coefficients $S(a, br; q)$. Combining the standard composite Kloosterman bound with a two-dimensional Erdős–Turán–Koksma inequality and partial summation gives the source-specific estimate

$$\boxed{\Delta_q(u, v) \ll q^{1/2+o(1)}\mathcal{B}(u)\mathcal{B}(v).} \tag{42}$$

For the actual nuclear smooth weights, $\mathcal{B}(u), \mathcal{B}(v) \ll (\log X)^{O(1)}$.

Status note G.2. Equation (42) is an internal analytic derivation for the proper smooth source. A formal proof assistant development should separate the finite Fourier identities from the external Kloosterman/ETK discrepancy input rather than encode the latter as an axiom.

G.4.3 Fixed-power margin

For a nonprincipal ratio character $\eta = \psi\overline{\chi} \neq 1$, the constant term m_q cancels, and the cross-character convolution operator has norm

$$\|T_{\text{off}}\|_{\text{op}} = \phi(q)\Delta_q(u,v) \ll q^{3/2}(\log X)^{O(1)}.$$

The natural uncentered (h,k) mass is

$$HK \asymp (Q/Y)^2.$$

Hence the ratio is

$$\frac{q^{3/2}}{HK} \asymp \frac{Y^2}{Q^{1/2}} = X^{2/9-\omega/2}.$$

At $\omega \geq 13/18$,

$$2/9 - \omega/2 \leq -5/36.$$

Thus

$$\boxed{\text{same-}q \text{ cross-character contribution} \ll X^{-5/36+o(1)} \times \text{natural source energy.}} \tag{43}$$

This closes the proper smooth cross-character child, but not the character diagonal.

G.5 Cross-q no-wrap moving-multiplier theorem

Let

$$c = q_1 q_2, \qquad (q_1, q_2) = 1,$$

and for $D \in G_c$ define

$$T_c(D) = \sum_{\substack{|h| \leq H \\ |k| \leq K}} \alpha_h \beta_k S(Dh, k; c).$$

Assume

$$2HK < c. \tag{44}$$

In the actual regime

$$H, K \asymp Q/Y, \qquad c \asymp Q^2,$$

so (44) holds for large X.

Character diagonalization gives

$$T_c(D) = \frac{1}{\phi(c)} \sum_{\chi \bmod c} \tau_c(\chi)^2 \overline{\chi}(D) A_\chi B_\chi,$$

with

$$A_\chi = \sum_h \alpha_h \overline{\chi}(h), \qquad B_\chi = \sum_k \beta_k \overline{\chi}(k).$$

Orthogonality in D yields

$$\sum_{D \in G_c} |T_c(D)|^2 = \frac{1}{\phi(c)} \sum_{\chi \bmod c} |\tau_c(\chi)|^4 |A_\chi|^2 |B_\chi|^2.$$

The character fourth moment expands into

$$\phi(c) \sum_{h_1 k_1 \equiv h_2 k_2 \pmod{c}} \alpha_{h_1} \beta_{k_1} \overline{\alpha_{h_2} \beta_{k_2}}.$$

By (44), the congruence lifts to the integer equality

$$h_1k_1 = h_2k_2.$$

The remaining multiplicative-convolution energy is bounded by divisor multiplicity:

$$\sum_n \left| \sum_{hk=n} \alpha_h \beta_k \right|^2 \ll c^{o(1)} \, \|\alpha\|_2^2 \, \|\beta\|_2^2 .$$

Together with the standard Gauss-sum bound,

$$\boxed{\sum_{D \in G_c} |T_c(D)|^2 \ll c^{2+o(1)} \, \|\alpha\|_2^2 \, \|\beta\|_2^2 .} \tag{45}$$

Consequently, for any moving-source coefficient $A(D)$,

$$\boxed{\left| \sum_{D \in G_c} A(D) T_c(D) \right| \ll c^{1+o(1)} \, \|A\|_2 \, \|\alpha\|_2 \, \|\beta\|_2 .} \tag{46}$$

Remark G.3. The analytic kernel in (46) is therefore controlled without a fixed-multiplier-to-family promotion. The remaining issue is the physical CRT push-forward norm of $A(D)$.

G.6 A route that can be retired

A direct completion in the whole products C, a formally gives

$$\sum_{C,a} F(C)G(a)\mathbf{1}_{Ca \equiv -2 \pmod q} = \frac{1}{q^2} \sum_{h,k \bmod q} \widehat{F}_q(h)\widehat{G}_q(k) S(h, -2k; q).$$

However, F and G are sparse divisor convolutions, not interval-smooth weights, so their Fourier transforms need not localize to short dual intervals.

Even under the counterfactual assumption of such localization, at the endpoint

$$q = X^{13/18}, \qquad C = X^{5/9}, \qquad a = X^{4/9},$$

one has

$$q/C = q^{3/13}, \qquad q/a = q^{5/13}.$$

The most favorable fixed-multiplier contraction supplied by the relevant bilinear Kloosterman range is still far smaller than the physical normalization requirement. Thus the whole-product route is nonclosing even before addressing the moving-source family. The slot-preserving (x_1, x_4) completion and the same-/cross-q sector split are the correct architecture.

G.7 The exact remaining closure system

The preceding analysis removes the need for a mysterious generic moving-multiplier theorem. The remaining obligations are source-exact.

G.7.1 Same-q character diagonal

For $\eta = \psi\overline{\chi} = 1$, the centred modular-hyperbola discrepancy does not provide cancellation. The required estimate has the schematic form

$$\sum_{q \sim Q} \frac{|\beta(q)|^2 Y^4}{q^4 \phi(q)^2} HK \sum_{\chi \bmod q} |\tau_q(\chi)|^4 |A_{\nu,q}(\chi)|^2 \ll_A \frac{X^2Y^2}{Q^2} (\log X)^{-2A}.$$

This must be proved using the literal seven-box source energy, the exact normalization of $\beta(q)$, and the existing residue/JQ7 router. It is not a consequence of the cross-character estimate.

G.7.2 Cross-q physical source spread

The analytic operator is controlled by (46). What remains is to prove for the physical CRT push-forward

$$(B_1, B_2) \longmapsto \Theta_{12}$$

a source norm of the form

$$\boxed{\|A_{q_1,q_2}\|_2 \le Y^{-1+o(1)} \|A_{q_1,q_2}\|_1\,,} \tag{47}$$

or any stronger equivalent estimate after restoring the full prefactor. This statement must include all labelled multiplicities and shared-prime strata.

G.7.3 Source-exhaustive Full-Nine/Ford reassembly

Closing the QK56 parent does not by itself prove that all balanced Full-Nine packets have been covered. The source compiler must establish an exhaustive packet identity and then map the actual generated Ford–Maynard grouped Type-II instances into Gate-1A, Gate-1B, or explicitly controlled exceptional packets.

A parallel source-definition obligation remains for the shifted four-cycle branch. Existing archives contain the four-cycle matrix/discriminant algebra, but not the literal pre-support-enlargement TT^* physical source. Before a source-linked character theorem is well posed one must recover:

$$\begin{aligned}
&B_1, \ldots, B_4, \qquad \Theta(B_i, B_{i+1}),\\
&h_i\text{-ranges and weights}, \qquad \text{prefactor/target},\\
&\text{degenerate strata}, \qquad \text{labelled multiplicity}.
\end{aligned}$$

G.8 Minimal honest dependency graph

The current Gate–1B closure path is therefore:

$$\begin{gathered}
\text{same-}q\text{ diagonal source energy}\\
+\text{cross-}q\text{ CRT source spread}\\
+\text{literal S1/S2/source normalizations}\\
+\text{finite shared-prime/nonunit/collision routing}\\
\downarrow\\
\text{QK56 full covariance closed}\\[4pt]
+\text{actual switched nine-coordinate dictionary}\\
+\text{literal shifted } TT^* \text{ source}\\
+\text{source-linked shifted child if required}\\
\downarrow\\
\text{Gate 1B closed}\\[4pt]
+\text{source-exhaustive Ford–Maynard packet compiler}\\
\downarrow\\
\text{downstream sieve rerun.}
\end{gathered}$$

G.9 Literature positioning

The determinant-character bounds of Fouvry–Shparlinski [FS22] and the monomial/reciprocal-monomial character-sum technology of Shparlinski [Shp13] are natural candidates only after a literal shifted-source dictionary is recovered. Bourgain–Garaev [BG12] provide reciprocal-set and multilinear Kloosterman technology that may become relevant to an explicit reciprocal product dictionary. Blomer–Pascadi [BP26] provide strong fixed-modulus bilinear Kloosterman capacity, but their theorem does not by itself supply the physical source push-forward norm in (47). Thus the current first blockers are source-energy and source-definition problems, not the absence of another generic fixed-multiplier theorem.

G.10 Status before the high-conductor signed-parent correction

The following ledger records the positive children established before Late update IV; it is not the controlling high-conductor status. The ledger is:

native first-TT^* sector law	proved internally,
same-q smooth modular-hyperbola discrepancy	analytic pass on proper smooth sector,
same-q cross-character Gram	analytic pass,
cross-q no-wrap moving-D L^2	proved at operator level,
same-q diagonal source energy	open,
cross-q physical source spread	open,
actual switched c_9 dictionary	source open,
shifted literal TT^* source	source open,
Gate 1B	**open**,
Full Type II / twin-prime infinitude	**not proved**.

H Late update IV: high-conductor complement sign recovery and common-conductor covariance

This section records the first high-conductor signed-parent correction. It is retained for audit history, but its final claim that the common-conductor covariance is the controlling atomic open is superseded by Late update V. It records only identities and reductions supported by the current source derivation. In particular, no application of Pascadi's Proposition 3.8 or of Blomer–Pascadi is claimed here. The principal architectural correction is that a repair which genuinely uses the Möbius sign of the modulus coefficient cannot close the same-q high-conductor sector in isolation.

H.1 Banked low/mid-conductor material

We retain the previously audited same-q collision dictionary and the low/mid-conductor same-q estimate. The latter still carries the source-audit requirement that the primitive large-sieve transcription use the correct $1/\phi(c)$ weighting and the actual common coefficient sequence. These children are not reopened below.

The native proper QK source remains

$$\sum_{C_5,x_2,x_3,q} \Gamma_5(C_5)f_2(x_2)f_3(x_3)\beta(q)\frac{Y^2}{q^2}\sum_{h,k}\Psi_{C_5,x_2,x_3,q}(h,k)S(hk\overline{B},\pm 2;q), \qquad B = C_5x_2x_3 \asymp Y^7,$$

with $|h|,|k| \ll q/Y$. The high-conductor correction below undoes only the outermost Cauchy/character-L^2 step needed to recover the signed parent; it does not replace the physical source by a new model.

H.2 Primitive conductor complement and exact sign factorisation

Write

$$q = cr, \qquad c \asymp C, \qquad C > C_* := (QY^{11})^{1/3}.$$

Then

$$r \ll \frac{Q}{C} \le \frac{Q}{C_*} = \left(\frac{Q^2}{Y^{11}}\right)^{1/3} \ll Y^{5/3+o(1)},$$

since $Q < Y^8$. The physical prime slot satisfies $p > V = Y^{5/2-o(1)}$, hence $p \nmid r$ in the residual band and the physical prime belongs to the primitive conductor c. Writing $c = pc_0$ and $d = c_0 r$, the literal high-conductor source factorisation is

$$\boxed{\beta(cr) = \mu(r)\, \varrho_{C,r}(c),} \tag{48}$$

where

$$\varrho_{C,r}(c) = \sum_{\substack{pc_0 = c \\ p \text{ and } c_0 r \text{ satisfy the physical boxes}}} \mu(c_0) \log p.$$

On the clean squarefree cell, $\mu(c_0) = -\mu(c)$ and therefore

$$\boxed{\beta(cr) = -\mu(c)\mu(r) L_{D,P}(cr),} \tag{49}$$

with

$$L_{D,P}(cr) = \sum_{\substack{p|c,\ p\in\mathcal{P} \\ (c/p)r\in\mathcal{D}}} \log p \ge 0.$$

The selector $L_{D,P}(cr)$ is genuinely joint in (c, r); no tensorisation $L_1(c)L_2(r)$ is used or claimed. We record

$$\boxed{\texttt{BETA-HICOND-COMPLEMENT-FACTOR45: internal proof.}}$$

H.3 Induced Gauss square and survival of the complement sign

Let $\chi \pmod{cr}$ be induced by a primitive character $\chi^* \pmod{c}$. On the proper squarefree sector,

$$\tau_{cr}(\chi) = \mu(r)\chi^*(r)\tau_c(\chi^*). \tag{50}$$

The QK character expansion contains the square of the Gauss sum, so

$$\tau_{cr}(\chi)^2 = \chi^*(r)^2 \tau_c(\chi^*)^2.$$

Combining this with (48) gives the exact identity

$$\boxed{\beta(cr)\tau_{cr}(\chi)^2 = \mu(r)\varrho_{C,r}(c)\chi^*(r)^2\tau_c(\chi^*)^2.} \tag{51}$$

Thus one factor $\mu(r)$ survives in the QK square-Gauss parent. This is the key distinction from the earlier one-Gauss expression, in which the induced Gauss factor spends the short-cofactor sign. We record

$$\boxed{\texttt{QK-GAUSS-SQUARE-SIGN-SURVIVES45: internal proof.}}$$

H.4 Exact signed parent before the sign-killing Cauchy step

For the proper unit sector define

$$\mathcal{A}_{c,r,\chi^*}(h,k) := \sum_{\substack{B \asymp Y^7 \\ (B,cr)=1}} \mathcal{G}_{cr}(B; h, k)\chi^*(B),$$

where $\mathcal{G}_{cr}$ contains the literal factors $\Gamma_5(C_5)$, $f_2(x_2)$, $f_3(x_3)$, $\Psi_{C_5,x_2,x_3,cr}(h,k)$ and $B = C_5x_2x_3$. Substituting (48) and (51) into the one-copy character formula yields

$$\boxed{\begin{aligned}\mathcal{S}_{\rm hi}^{\rm preC} = Y^2 \sum_{\substack{c>C_* \\ cr\asymp Q \\ (c,r)=1}} \frac{\mu(r)\varrho_{C,r}(c)}{c^2r^2\phi(cr)} \sideset{}{^*}\sum_{\chi^* \bmod c} \tau_c(\chi^*)^2\chi^*(r)^2 \\ \times \sum_{h,k} \overline{\chi^*}(bhk)\mathcal{A}_{c,r,\chi^*}(h,k),\end{aligned}} \tag{52}$$

with $b = \pm 2$ according to the fixed source convention. On the clean squarefree cell, (49) gives the equivalent form with $-\mu(c)\mu(r)L_{D,P}(cr)$ in the numerator. Axes, non-unit, repeated-prime and shared-prime strata remain in their existing routers. Hence

`HICOND-SIGNED-PARENT-RECOVERED: internal proof.`

H.5 Why the old SHORT-RJ pre-Cauchy geometry is false

The positive collision $B_2 - B_1 = jc$ arises only after opening $\sum_{\chi^*} |\mathcal{A}_{c,r,\chi^*}|^2$ and applying character orthogonality. That same step replaces the linear source coefficient $\beta(cr)$ by $|\beta(cr)|^2$ and therefore destroys the complement Möbius sign. Conversely, after undoing that Cauchy step, the signed parent (52) contains one B variable and no shift j. Consequently the numerical identity $R_*J_* = H^{1/3}$ belongs to the post-Cauchy positive collision geometry and cannot simultaneously be used as a pre-Cauchy signed entropy resource. We therefore record the death certificate

`SHORT-RJ-PRECAUCHY-SIMULTANEOUS-DICTIONARY45:  false.`

H.6 High conductor must recombine same-q and common-conductor cross-q

For any family T_q,

$$\left|\sum_q \beta(q)T_q\right|^2 = \sum_q |\beta(q)|^2|T_q|^2 + \sum_{q_1\neq q_2} \beta(q_1)\overline{\beta(q_2)}T_{q_1}\overline{T_{q_2}}.$$

The same-q part only sees $|\beta(q)|^2$ and therefore cannot use the Möbius sign. Any high-conductor argument which genuinely uses the recovered sign must retain cross-q covariance. In the common-conductor sector $q_i = cr_i$, the sign carrier is $\mu(r_1)\mu(r_2)$ and the same-q case $r_1 = r_2$ is merely its positive diagonal. Thus

`HICOND-COMMON-CONDUCTOR-RECOMBINATION45:  internal proof.`

This supersedes the earlier sequential plan "close same-q, then cross-q" for the high-conductor floor.

H.7 Square-complement collision and the no-wrap threshold

If one now takes a character L^2 step while retaining the (r_1, r_2) coherence, the primitive-character projector gives, on the proper unit sector,

$$\boxed{\sideset{}{^*}\sum_{\chi^* \bmod c} \chi^*(r_1^2B_1)\overline{\chi^*(r_2^2B_2)} = \sum_{d|(c,r_1^2B_1-r_2^2B_2)} \phi(d)\mu(c/d).} \tag{53}$$

Hence the top $d = c$ stratum has the collision

$$\boxed{r_1^2 B_1 - r_2^2 B_2 = j_{\mathrm{sig}} c,} \tag{54}$$

not $B_1 - B_2 = jc$. If $r_i \asymp R = Q/C$ and $B_i \asymp Y^7$, then

$$|j_{\mathrm{sig}}| \ll \frac{R^2 Y^7}{C} = \frac{Q^2 Y^7}{C^3}.$$

The corresponding no-wrap threshold is

$$\boxed{C_{\mathrm{nw}} := (Q^2 Y^7)^{1/3}.} \tag{55}$$

Moreover

$$C_{\mathrm{nw}} < Q \quad \Longleftrightarrow \quad Q > Y^7,$$

so the global phase boundary $\omega = 7/9$ reappears for a precise reason: it is the point at which a signed square-complement no-wrap slice exists. For $C > C_{\mathrm{nw}}$, on the fully proper cross-coprime squarefree stratum, the top collision reduces to $r_1^2 B_1 = r_2^2 B_2$; under $(B_1 B_2, r_1 r_2) = 1$, unique factorisation forces $r_1 = r_2$ and $B_1 = B_2$. Thus the top projector is positive diagonal in the upper no-wrap slice; any cancellation must come from a proper-divisor projector or a separately routed shared-prime stratum.

H.8 Source-stage firewall against a direct MAM45/HFMV identification

The recovered parent (52) already lies after 2-dimensional Poisson completion, Kloosterman completion and a primitive-conductor character projection. Undoing the final Cauchy step does not invert these earlier transforms or remove the analytic selector $\mathrm{cond}(\chi) = c > C_*$. Therefore the present source does not supply a literal operator identity from the high-conductor QK parent to the pre-completion MAM45 or HFMV parents. We record

$$\boxed{\texttt{DIRECT HICOND-QK-PARENT} \to \texttt{MAM45/HFMV} : \text{source-stage mismatch.}}$$

A global MAM45/HFMV theorem could of course control the full physical parent and hence its high-conductor residual after subtraction of already-controlled pieces; that is a global theorem landing, not a local dictionary.

H.9 The exact atomic high-conductor covariance

Put

$$\mathcal{Z}_{c,r,\chi}(h,k) := \overline{\chi}(bhk)\mathcal{A}_{c,r,\chi}(h,k).$$

The exact common-conductor covariance is

$$\boxed{\begin{aligned} \mathcal{C}_{\mathrm{cc}}^{\mathrm{hi}} := Y^4 \sum_{c > C_*} \sum_{\substack{r_1, r_2 \\ cr_i \asymp Q}} &\frac{\mu(r_1)\mu(r_2)\varrho_{C,r_1}(c)\overline{\varrho_{C,r_2}(c)}}{c^4 r_1^2 r_2^2 \phi(cr_1)\phi(cr_2)} \\ \times \sideset{}{^*}\sum_{\chi^* \bmod c} &|\tau_c(\chi^*)|^4 \chi^*\left(\frac{r_1^2}{r_2^2}\right) \sum_{h,k} \mathcal{Z}_{c,r_1,\chi^*}(h,k)\overline{\mathcal{Z}_{c,r_2,\chi^*}(h,k)}. \end{aligned}} \tag{56}$$

The required high-conductor theorem is

$$\boxed{\mathcal{C}_{\mathrm{cc}}^{\mathrm{hi}} \ll_A \frac{X^2 Y^2}{Q^2} (\log X)^{-A}.} \tag{57}$$

This single parent contains the positive same-q floor $r_1 = r_2$, the cross-q terms which carry the only available complement Möbius sign, the primitive-projector collision (54), and the actual five-defect QK coefficients. It is therefore the controlling atomic high-conductor open.

H.10 Square-complement adapter: candidate only

Equation (51) shows that the complement phase is quadratic in r. Formally, if a later source-preserving Kloosterman transcription produces $\overline{r}^{\,2}$, the positive-integer map $s = r^2$ is injective and preserves both ℓ^1 and ℓ^2 norms of the outer weights. In the high-conductor range,

$$\frac{R^2}{C} = \frac{Q^2}{C^3} < \frac{Q}{Y^{11}} \ll Y^{-3+o(1)}.$$

This makes $s = r^2$ a genuinely short outer variable relative to the primitive conductor. It motivates a fresh audit against Pascadi's variable-family Kloosterman machinery [Pas25], but no application is claimed here. Before such an invocation one must prove the literal $\overline{r}^{\,2}$ Kloosterman slot, place the joint selector $\varrho_{C,r}(c)$ in a legal coefficient architecture, expose the genuine multiplicative smooth slots, verify the relevant exceptional-spectrum hypothesis, and recompute every geometric term after replacing the outer range R by R^2. These are open source/analytic tasks.

H.11 Controlling Gate–1B ledger after the correction

The current high-conductor ledger is

same-q collision dictionary	banked,
same-q low/mid conductor	banked; weighting audit tag retained,
$q = cr$ complement geometry	proved internally,
$\beta(cr) = \mu(r)\varrho_{C,r}(c)$	proved internally,
$\tau_{cr}(\chi) = \mu(r)\chi^*(r)\tau_c(\chi^*)$	proved internally,
$\beta(cr)\tau_{cr}(\chi)^2$ retains $\mu(r)$	proved internally,
signed pre-Cauchy parent	recovered,
old SHORT-RJ pre-Cauchy dictionary	**false / retracted**,
correct signed collision	$r_1^2B_1 - r_2^2B_2 = j_{\mathrm{sig}}c$,
$C_{\mathrm{nw}} = (Q^2Y^7)^{1/3}$	proved threshold,
same-q high conductor as standalone sign repair	invalid architecture,
common-conductor recombination	proved structurally,
direct high-QK to MAM45/HFMV	source-stage mismatch,
$\mathcal{C}^{\mathrm{hi}}_{\mathrm{cc}}$ target	**first high-conductor analytic open**,
square-complement Pascadi adapter	candidate/open,
Gate 1B	**open**.

Accordingly this section replaces the old SHORT-RJ and standalone same-q high-conductor architectures. The subsequent square-complement audit in Late update V goes further: it proves that the common-conductor block is not the full parity-sensitive parent and retires the direct Pascadi-3.8 closure route.

I Late update V: square-complement death certificate and the near-primitive first open

This section is the controlling Gate–1B update for the high-conductor range. It records the hostile audit of the common-conductor square-complement adapter. The outcome is a genuine DAG repair rather than a positive Pascadi splice: the short-r square norm and a primitive-Kloosterman inversion are exact, but the naive r^2 dictionary is false, both legal Proposition 3.8 lanes are power-nonclosing, and the common-conductor covariance has already squared away the conductor Möbius sign. The unavoidable first analytic child is therefore the near-primitive full-conductor five-defect signed packet.

I.1 Exact short-complement square norm

Put

$$\lambda_{c,r} := \frac{\mu(r)\varrho_{C,r}(c)}{r^2\phi(cr)}.$$

Using

$$\chi^*\left(\frac{r_1^2}{r_2^2}\right) = \chi^*(r_1)^2\overline{\chi^*(r_2)^2},$$

the common-conductor covariance (56) is exactly

$$\boxed{\mathcal{C}_{\mathrm{cc}}^{\mathrm{hi}} = Y^4 \sum_{c>C_*} \frac{1}{c^4} \sideset{}{^*}\sum_{\chi^* \bmod c} |\tau_c(\chi^*)|^4 \sum_{h,k} \left| \sum_{\substack{r\\ cr\asymp Q}} \lambda_{c,r}\chi^*(r)^2 \mathcal{Z}_{c,r,\chi^*}(h,k) \right|^2 .} \tag{58}$$

No estimate is used in (58); it is simply a reassembly of the r_1, r_2 covariance into a short-complement square norm. We record

$$\boxed{\texttt{COMMON-CONDUCTOR-SHORT-r-SQUARE-NORM45: internal proof.}}$$

I.2 The genuinely short quantity and the physical aggregate variable

With $R = Q/C$ and $C > C_* = (QY^{11})^{1/3}$,

$$\frac{R^2}{C} = \frac{Q^2}{C^3} < \frac{Q}{Y^{11}} < Y^{-3+o(1)}, \qquad R^2 \ll CY^{-3+o(1)}. \tag{59}$$

Thus r^2 by itself is a very short integer variable relative to the primitive conductor. The decisive correction is that the physical Kloosterman inverse variable is not r^2 alone. Since $B \asymp Y^7$, the aggregate scale is

$$\frac{R^2Y^7}{C} = \frac{Q^2Y^7}{C^3}, \tag{60}$$

which is < 1 only after $C > C_{\mathrm{nw}} = (Q^2Y^7)^{1/3}$. Hence (59) does not supply a free short-variable theorem throughout the high-conductor range.

I.3 Primitive-Kloosterman inversion

For squarefree c define the primitive kernel

$$\mathfrak{P}_c(x) := \sideset{}{^*}\sum_{\chi \bmod c} \tau_c(\chi)^2\overline{\chi}(x).$$

Decomposing every character modulo c by its primitive conductor and using the induced-Gauss identity gives

$$\phi(c)S(1,x;c) = \sum_{d|c} \mathfrak{P}_d\Big(x\overline{(c/d)^2}; d\Big).$$

Möbius inversion on the divisor lattice therefore yields the exact identity

$$\boxed{\mathfrak{P}_c(x) = \sum_{d|c} \mu(c/d)\phi(d)S\Big(1, x\overline{(c/d)^2}; d\Big).} \tag{61}$$

For $c = p$ prime this becomes $\mathfrak{P}_p(x) = (p-1)S(1, x; p) - 1$, as expected from "all characters minus principal". Substituting the physical $x = bhk\overline{r^2B}$ and writing $c = de$ gives

$$\boxed{\mathfrak{P}_c\Big(bhk\overline{r^2B}\Big) = \sum_{de=c} \mu(e)\phi(d)S\Big(hk\overline{(er)^2B}, b; d\Big).} \tag{62}$$

Thus even the top $d = c$ term contains the inverse of r^2B, not the inverse of r^2 alone. We record

$\boxed{\texttt{PRIMITIVE-KLOOSTERMAN-INVERSION45:\ \ internal proof.}}$

I.4 Why the naive Pascadi-3.8 square-complement dictionary is false

Pascadi's Proposition 3.8 is a theorem for a kernel of the schematic form

$$S(m_1m_2\bar{\mathfrak{r}}, \pm n; \mathfrak{s}\mathfrak{c})$$

with two genuine multiplicative-convolution slots m_1, m_2 and an arithmetic sequence subject to its Assumption 3.2 [Pas25]. The physical top kernel in (62) is

$$\boxed{S(hk\overline{r^2B}, b; c), \qquad B = C_5x_2x_3.}$$

The two multiplicative slots are already forced to be $m_1 = h$ and $m_2 = k$. Consequently the five defects and two model factors cannot be reassigned to those slots without changing the source. There are only two legal broad arrangements.

First, scaling B into the second Kloosterman argument gives

$$S(hk\overline{r^2B}, b; c) = S(hk\overline{r^2}, b\overline{B}; c),$$

so one may take $\mathfrak{r} = r^2$ and encode $n \equiv b\overline{B} \pmod{c}$ in an inverse-residue coefficient sequence. This is structurally legal, and the general-sequence form of Assumption 3.2 only inserts the positive norm

$$A^2_{r^2,c} = \sum_n |a_{n,r^2,c}|^2.$$

The theorem therefore requires, rather than proves, the physical inverse-residue spread. The literal parameter audit in the present source yields at best

$$|\mathcal{S}_{\mathrm{hi}}(C)| \ll X^{o(1)}CQ^{1/2}A_C. \tag{63}$$

In the residue-injective upper range $C, Q \ge Y^7$, the natural value $A_C = Y^{7/2+o(1)}$ leaves a deficit at least Y^5. Hence

$\boxed{\texttt{LEGAL-P38-INVERSE-RESIDUE-LANE45:\ \ power-nonclosing.}}$

Second, one may put the entire product r^2B into the outer inverse variable. Then

$$\mathfrak{R} \asymp R^2Y^7 = \frac{Q^2Y^7}{C^2},$$

and the resulting geometric factor and source norm are even larger; the source audit gives an output of size

$$Y^{23/2+o(1)}\sqrt{CQ},$$

which is far above the physical target Y^9. Hence

`LEGAL-P38-PRODUCT-OUTER-LANE45`: **EVEN-MORE-NONCLOSING**.

The common-conductor Pascadi closure route is therefore retired. This is a capacity and source-decomposition death certificate; it is not a claim that Proposition 3.8 is incorrect.

I.5 The deeper sign loss inside the common-conductor covariance

On the clean cell,

$$\beta(cr) = -\mu(c)\mu(r)L_{D,P}(cr).$$

But within a common-conductor covariance,

$$\beta(cr_1)\overline{\beta(cr_2)} = \mu(c)^2\mu(r_1)\mu(r_2)L_{D,P}(cr_1)L_{D,P}(cr_2),$$

so the conductor sign $\mu(c)$ has already been squared away. The block only retains the complement pair $\mu(r_1)\mu(r_2)$. In particular the near-primitive slice $r_1 = r_2 = 1$ contains no complement sign at all. Thus

$$\boxed{\texttt{COMMON-CONDUCTOR-ONLY-SIGN-REPAIR45:\ \ incomplete}.}$$

The common-conductor square norm remains a legitimate secondary $r > 1$ child, but it cannot be treated as the full parity-breaking high-conductor parent.

I.6 Representation-loop interpretation

Equation (61) also explains why the square-complement route does not evade the earlier fixed-modulus death certificate. If the primitive conductor c is kept fixed, the quadratic complement phase remains visible but there is no single complete Kloosterman sum. If one expands the primitive projector into complete Kloosterman sums, one must sum over $d \mid c$ and replace r by er with $e = c/d$; summing all primitive conductors reconstructs the original full-character QK5 source. The source energy therefore migrates into the inverse-residue norm or the outer-weight norm rather than disappearing.

I.7 The unavoidable near-primitive signed child

Every r^2-based complement argument must confront $r = 1$. After a fixed polylogarithmic nuclear separation, one near-primitive source packet has the form

$$\boxed{\begin{aligned}\mathcal{S}_{\mathrm{NP}}^{(\nu)} = -Y^2\sum_{c\asymp Q}\frac{\mu(c)L_{D,P}(c)}{c^2\phi(c)}\sum_{\chi^*\bmod c}^{*}\tau_c(\chi^*)^2\prod_{i=1}^{5}D_{i,\nu,c}(\chi^*)F_{2,\nu,c}(\chi^*)F_{3,\nu,c}(\chi^*)\\ \times\overline{H_{\nu,c}(\chi^*)K_{\nu,c}(\chi^*)}.\end{aligned}} \tag{64}$$

Here the conductor Möbius sign $\mu(c)$ and all five centred defect factors remain linear; there is no complement variable available for square-complement averaging. The first unavoidable high-conductor theorem is therefore

$$\boxed{\mathcal{S}_{\mathrm{NP}}^{(\nu)} \ll_A X(\log X)^{-A}.} \tag{65}$$

Equivalently, at the pre-completion source stage one must bound the signed native determinant packet

$$\boxed{\sum_{\substack{B,x_1,x_4,c,\ell\\ Bx_1x_4-c\ell=-2}}\Gamma_5(C_5)f_2(x_2)f_3(x_3)\mu(c)L_{D,P}(c)W(\cdots) \ll_A X(\log X)^{-A}.} \tag{66}$$

For $r = 1$, the top $d = c$ term of (61) is precisely $\phi(c)S(hk\overline{B}, b; c)$, the original QK5 complete kernel. Thus the finite/source inversion back to the native determinant shell $Bx_1x_4 - c\ell = -2$, while preserving $\beta(c) = -\mu(c)L_{D,P}(c)$, is an exact source adapter. It is not an estimate. We record

$$\boxed{\texttt{NEARPRIM-TOP-KLOOSTERMAN-TO-SIGNED-QK5/MAM45-ADAPTER: pass}.}$$

I.8 Revised Gate–1B high-conductor DAG

The controlling high-conductor path is now

$$
\boxed{
\begin{array}{c}
\text{low/mid conductor bank} + \;\text{cross-character bank} + \;\text{cross-}q\text{ no-wrap operator bank} \\
\downarrow \\
\textbf{HIGH CONDUCTOR} \quad \begin{cases} r = 1 : & \boxed{\texttt{NEARPRIM-FULL-c-FIVEDEFECT-SIGNED45}}, \\ r > 1 : & \text{square-character complement family}, \end{cases} \\
\downarrow \\
\text{full } (c_1, r_1) \times (c_2, r_2) \text{ source reassembly} \\
\downarrow \\
\text{QK56 parent} \\
\downarrow \\
\text{exhaustiveness / shifted-child test} \\
\downarrow \\
\texttt{GATE1B CLOSED}.
\end{array}
}
$$

The common-conductor covariance is retained only as a secondary $r > 1$ child. The present first analytic open is NEARPRIM-FULL-c-FIVEDEFECT-SIGNED45.

I.9 Controlling ledger after the square-complement audit

common-conductor short-r square norm	proved exactly,
$R^2/C \ll Y^{-3+o(1)}$	proved,
primitive-Kloosterman inversion	proved exactly,
actual inverse variable	r^2B, not r^2,
naive $r^2 \to$ Pascadi-3.8 dictionary	**false**,
legal inverse-residue P3.8 lane	**power-nonclosing**,
legal product-outer P3.8 lane	**even-more-nonclosing**,
common-conductor $\mu(c)$	squared away,
$r = 1$ near-primitive child	**unresolved signed floor**,
common-conductor P3.8 closure	**retired**,
new first high-conductor open	$\boxed{\texttt{NEARPRIM-FULL-c-FIVEDEFECT-SIGNED45}}$,
Gate 1B	**open**.

J Working missing-literature and source-definition bank

Status note J.1 (Editorial working ledger)**.** This appendix is a working retrieval ledger rather than part of the proof of any theorem above. It is retained so that source and literature gaps can be cleared systematically in a later final audit. In particular, a missing *programme source definition* is not to be relabelled as a missing published theorem.

J.1 ML/SOURCE-1: Gate-1B physical-source definition gap

The current archive search recovers substantial finite compiler mathematics, but does not recover the two physical source definitions needed to instantiate the remaining Gate-1B compiler. We therefore bank a single item

$$\boxed{\texttt{ML/SOURCE-1 --- Gate1B physical-source definition gap}}.$$

This is principally a *source-definition* gap, not a missing-paper gap.

Recovered finite/source algebra. The supplied archives contain the switched operator and its exact multiplier reindexing, the $\mu * \Lambda$ divisor opening with ordered divisor-pair multiplicity retained, and the higher-prime-power/repeated-prime/generic switched partition. They also contain the conditional Q5 support identity: if the relevant coefficient is literally a Dirichlet convolution $c = a * b$, then the generic switched term expands onto the determinant shell

$$mn + 2 = dpr.$$

Likewise, the formal nine-coordinate ANOVA/distributive compiler and the finite packet-reassembly compiler are available. None of these finite identities, however, identifies the actual physical nine-coordinate coefficient.

Missing source definition 1: actual switched $R9$ cell. The first source-open item is

$$\boxed{\texttt{ACTUAL-SWITCHED-R9-CELL-DICTIONARY45}}.$$

The formal interface permits a relation of the form

$$c_9(t) = \kappa_j\,(\alpha_j * \beta_{9-j})(t) + E_j(t),$$

but the supplied source does not prove that the actual physical c_9 has this form. A source-safe dictionary must identify, for every actual packet,

- the literal sequences α_j and β_{9-j};
- labelled versus symmetrised source coordinates and their dyadic ranges;
- the Möbius, prime and model scalars;
- ordered-factorisation multiplicity and repeated-prime correction;
- smooth/Perron weights, the original prefactor and physical target;
- exceptional and degeneracy routing.

Until this dictionary exists, the exact nine-coordinate convolution compiler is only conditional on its source antecedent.

Missing source definition 2: shifted TT^* physical source. The second source-open item is

$$\boxed{\texttt{SHIFT-TTSTAR-LITERAL-SOURCE-PIN}}.$$

The archive contains four-cycle matrix, trace, determinant and discriminant algebra, and counterguards against treating Cauchy copies as independent physical multipliers. It does not contain the literal pre-support-enlargement TT^* source formula from which the cyclic multipliers should be derived. Before `SHIFT-SOURCE-LINKED-CHAR45` is a fully source-posed theorem, the source pin must specify

- the literal definitions and ranges of $B_1, \ldots, B_4$;
- the exact cyclic law $\Theta(B_i, B_{i+1})$;
- all h_i ranges and source weights;
- the prefactor, normalisation and physical target;
- the $D = 0$, non-unit, shared-prime, repeated-prime and collision strata;
- labelled tuple multiplicity and the exact equality/error across any support enlargement.

Why literature search cannot by itself close this item. Published determinant-character, trace-function and Kloosterman estimates can be matched only *after* the physical coefficient and multiplier family are defined. They cannot determine a private programme coefficient such as c_9, the variables B_i, the cyclic map $\Theta(B_i, B_{i+1})$, or the packet multiplicity. Ford–Maynard's published sieve framework likewise does not define these programme-specific objects [FM24]. Consequently the next action for this bank item is source archaeology or a literal source derivation, not broader theorem shopping.

Current status and later final-audit checklist. We retain

switched finite compiler	pass,
nine-coordinate convolution compiler	conditional pass,
finite Gate-1B reassembly compiler	pass,
actual switched $R9$ source dictionary	source open,
shifted TT^* physical source	source open,
Gate 1B	open.

When the final missing-literature/source audit is compiled, this item should be cleared only by one of the following: (i) an authoritative source file containing the missing literal definitions; (ii) a derivation from an earlier physical source which proves them exactly, including multiplicity and normalisation; or (iii) a proof that the corresponding packet is absent and therefore need not be routed. A published black box matching only the later schematic kernel is not sufficient.

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